

Outline (until the next coffee break)

- Lecture: Modeling of regulatory activity with sequence models
- Hands-on: Training a sequence-based model
- Discussion of pre-training and fine-tuning approaches
- Hands-on: Interpreting models with in-silico mutagenesis



Sequence modeling

ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Arjun Devadas

Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



Scope of this session

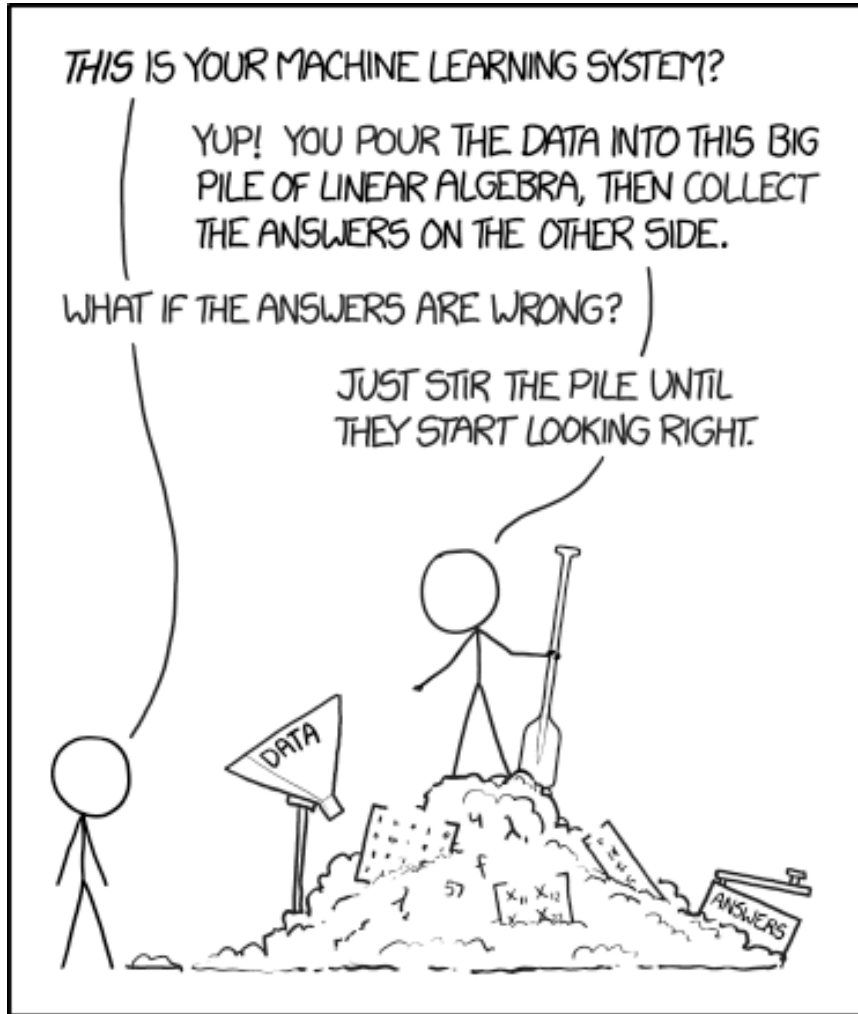
Goals

- Intuitive understanding of machine learning and basic sequence models
- Neural networks (NNs): What are they? and How they work?
- Example: **DeepSTARR**
- Model Interpretation

Questions we will cover:

- Why sequence-based modeling?
- How to leverage NNs alongside experimental data?
- What can we gain from NNs and how?

Machine learning 101



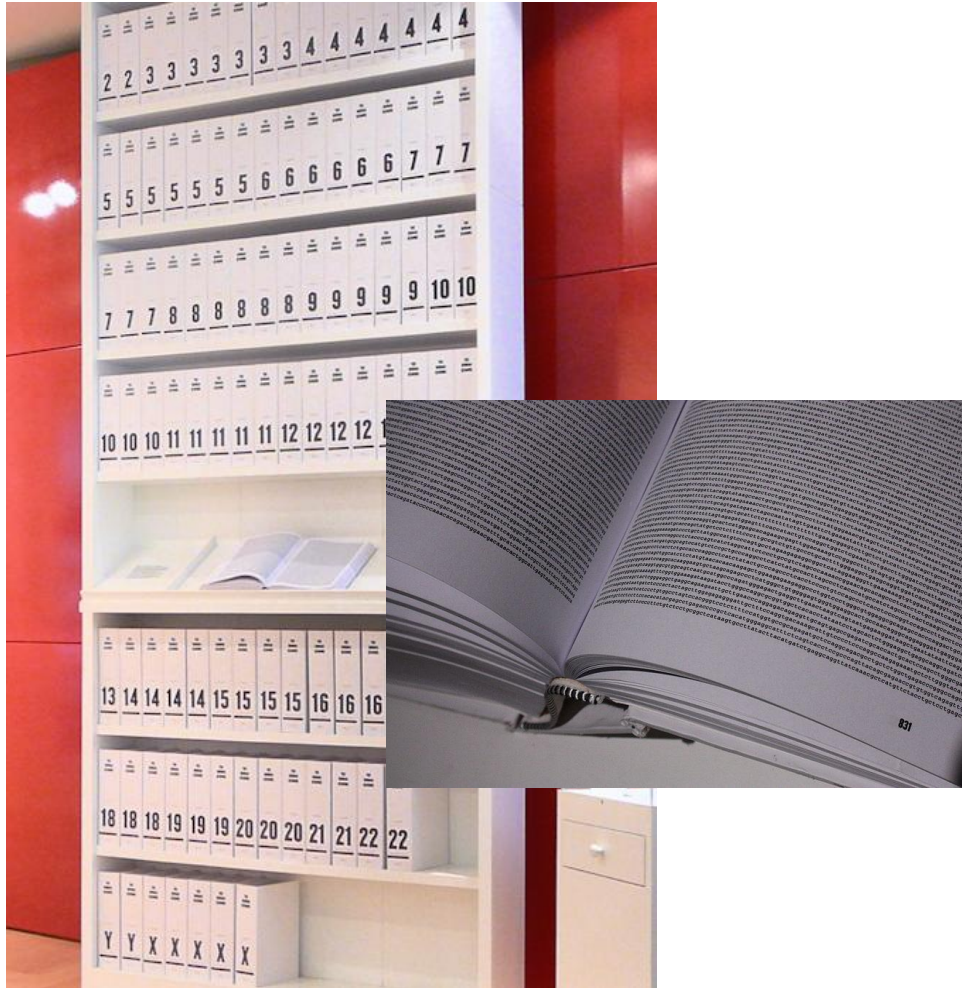
make computer systems ... learn
generalizations ... (without explicit
information) ... from patterns of data

<https://xkcd.com/1838/>

Basic terminology

- **Data – typically transformed to derive features**
- **Typically, a data set (features + output/response) is split into:**
 - **train and test set**, entries within these sets are called **examples**
 - sometimes an (independent) **validation set** is used, and the **test set serves for optimizing hyperparameters** (i.e., all kinds of tuning parameters to the model, like the number of parameters or training iterations)
- **Responses can be categorical (classification) or numerical (regression)**
- **Model training is either:**
 - Supervised: response is available, and the model is trained to produce the response
 - Semisupervised: proportion of training examples does not have a response
 - Unsupervised: clustering (e.g. kmeans, graphs/cliques, self-organizing maps)
- **Model performance is assessed by comparing predicted responses with original responses**
- **Models can be trained on surrogate data / correlated problems, recently called “zero-shot”**

Sequence modeling

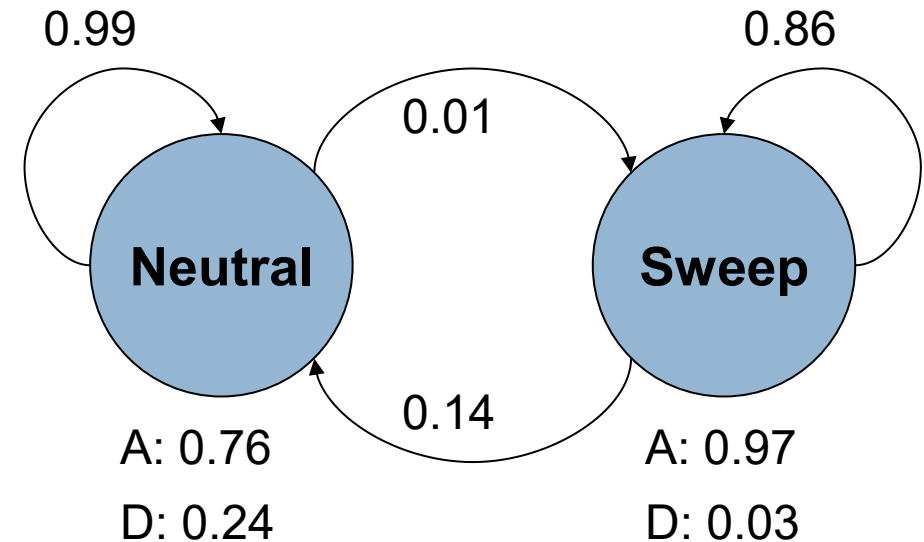


Printed Human Genome of the Wellcome Collection

CCTGGTGGAGAATTTAGTGACATTTGTTGGCTGCTTCTTAGGGAAGAAACCAAAGTCTGAC
TCTAACTCCACAGTATTTTTCCCCCTGGTGACTGACACAGCTGACCTAGAGCTAGAAGTCACC
TCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGAGAGGACATGGACCCTGAGGGC
CTCTCCACACAGACTGGGCCTCACCTCTCCACAGAGGAATTCAGCTCAGGAGTAGATGTAC
AGATGCCGGCACGATCTGCTTACAGCTTCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTTTT
TTTGGCTACTGAGCTCATCAAAGCTGGTGAGGACCCAGCTAAGGTTGTTTGCTGCTGCTGTG
AGCACTTGAGTAGTAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCCTCAAGGGGGCAGCCA
CAGCCCCTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGAC
ATTCTGATCCCAATCTCACTTAAGGATTAAGGGAGGGGGCAGTAGGGAGACAGGATAGCCAG
TCTTGTAATTTGAGACGGGCCCAGGATCCCTTATCCTGCAATCTGTCCTGCTGGACAGAGCA
CTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTGGGAGCAGTGACTGCAGACTCAG
AGGGTCTCTTCTCCTGGATCCTTATCTGGCTTCCACCCCAGGAAGCCAGGCCATTGCTACAC
CTGTGTGAGGCTCTCTAGGGAGATGGGGCTGGGGCTGGCCCTGCTTCCCTTCTGCATTTATC
AAGTGCCTGTGACAGGGAAGACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCA
GGTCTTGAAATAGTGAAGTTCAGTTTCTAAACCCCTCTGTGCCTCCTTCTTGCTCTCTGTAA
AATGCTGCGGTTGGAGAGATGGTGGATACAGGGCCTTTTAAAGTTGAATTTCCAGATGATTG
GTTTCCAAGCTTTCTGAAAAATGAGGCCCTTATCAATGTTGAAAAATAGGTAAATCTCATT
CTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTTGTAACCCAAAGTATAAAGGCT
TTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTACCTAGGCTGGATTGCAGTGGCATAA
TCACAGCTCACTGCAGCCTTGACCTCCCAGGCTCAGGCGACCTTCCTGCCTCAGCCTCCTGA
ATAGCTGGGATTATAGGTGTGCATCACTATGCCTAGTTAATTTTTGTATTTTTTGTAGAGAC
GGGGCTTCATCATGTTGCCCAAATGCTTGATGTGCTTATTTACATTCCATGCCTGTATCAA
AACATCTCATGTACCCCATACATGTATATACCTACTATGTATGCATAAAAATTTTTTAAAT
AATTTTTTAAATTAAAAAAATTCTGAGAATTTAAGCTTTTAAATCTCAGTTGAGTACTAAGTG
AATGGTAATTAAAGACTAAAATTACTACCAGTTGTCATATTTTAAATTGAATTTAAATTTAC
CATTGCAATCATAGAAGTCAACTCTTATTGAAAGCTTAGTCTGTGCCAAGCTCTATTCTAAA
ATGTATTGACCCTAAATACATTTAGATAATGTATACATTATCTAAATGTAATAAAAAAATA
ATAATTTTTGTTATTATTTCCATCCTATGGATGAAGCTGCTGAAACCCAAGTCCCCTACT
GGGTGGTGGCAACACTATGAGCTGTTGGGAAAATATACTAAATCTATTATTAAGTTTAAGT
AAAATAATTGAATATTATATAATTATATGAATTAATCAAATATAATTAAATATATTTCATAA
ATAAATATTATTAGAAATAATTCAAGGAGAAAAAAATCCCCAACCCAGAAAAACCTGAGG

A classic – Hidden Markov Models (HMMs)

- E.g. two state model of genomic regions under neutral evolution vs selection, based on sequence of single nucleotide polymorphisms in an ancestral (A) or derived (D) state (similar to “Occasionally dishonest casino” with fair vs loaded coin)
- HMMs can be used to:
 - Predict gene models, e.g. define Exon/Intron boundaries
 - Segment the genome based on sequence and/or sequence annotations (e.g. ChromHMM, Segway)



kmer representation

```
CCTGGTGGAGAATTTAGTGACATTTGTTGGCTGCTTCTTAGGGAAGAAACCAA  
GTCTGACTCTAACTCCACAGTATTTTCCCCCTGGTGACTGACACAGCTGACCTAG  
AGCTAGAAGTCACCTCATCTGACCCCTGGTTGCTGGCTCTCCACTTGACAATGGA  
GAGGACATGGACCCTGAGGGCCTCTCCACACAGACTGGGCCTCACCTCTCCACA  
GAGGAATTCAGCTCAGGAGTAGATGTACAGATGCCGGCACGATCTGCTTACAGCT  
TCTCATTGCAGGCATAGTCTTTTTTTTTTTTTTTTTTTGGCTACTGAGCTCATCA  
AAGCTGGTGAGGACCCAGCTAAGGTTGTTTGCTGCTGCTGTGAGCACTTGAGTAG  
TAATTGGGGTGTGGTGAGGGAGAAAACAAATGCCTCAAGGGGGCAGCCACAGCCC  
CTAACTCTAGGCCTGACAGAGAAGGCAGGGAGCTCGGCAGACAGGTGCGTGGAGA  
CATTCTGATCCCAATCTCACTTAAGGATTAAGGGAGGGGGCAGTAGGGAGACAGG  
ATAGCCAGTCTTGTAATTTGAGACGGGCCAGGATCCCTTATCCTGCAATCTGTC  
CTGCTGGACAGAGCACTGTCCTCAGCACAGAGGCAAGGTCCTCAATAAAGAGCTG  
GGAGCAGTGACTGCAGACTCAGAGGGTCTCTTCTCCTGGATCCTTATCTGGCTTC  
CACCCCAGGAAGCCAGGCCATTGCTACACCTGTGTGAGGCTCTCTAGGGAGATGG  
GGCTGGGGCTGGCCCTGCTTCCCTTCTGCATTTATCAAGTGCCTGTGACAGGGAA  
GACACCACAAGAGGGTCACCGACCTGAGGGCAGTGCAGTCCCAGGTCTTGAAATA  
GTGAAGTTTCAGTTTCTAAACCCCTCTGTGCCTCCTTCTTGCTCTCTGTAAAAATGC  
TGCGGTTGGAGAGATGGTGGATACAGGGCCTTTTAAGTTGAATTTCCAGATGAT  
TGGTTTCCAAGCTTTCTGAAAAATGAGGCCCTTATCAATGTTGAAAAATAGGTA  
AATCTCATTCTTTTTTATGACTGAACTTAAAGAATGTGATTGGATTGTTGTAAAC  
CCAAAGTATAAAGGCTTTTTTTTTTTTTTCTGAGACAAGGTCTCACTCTGTCACC  
TAGGCTGGATTGCAGTGGCATAATCACAGCTCACTGCAGCCTTGACCTCCCAGGC  
TCAGGCGACCTTCCTGCCTCAGCCTCCTGAATAGCTGGGATTATAGGTGTGCATC  
ACTATGCCTAGTTAATTTTTTGTATTTTTTGTAGAGACGGGGCTTCATCATGTTGC  
CCAAATGCTTGATGTGCTTATTTTACATTCATGCCTGTATCAAAACATCTCATG  
TACCCCATACATGTATATACCTACTATGTATGCATAAAAAATTTTTTAAATAATT  
TTTAAAAATTAAAAAATTTCTGAGAATTTAAGCTTTTAACTCAGTTGAGTACTAA  
GTGAATGGTAATTAAAGACTAAAAATTACTACCAGTTGTCATATTTTAAATTGAATT  
TAAAAATTTACCATTGCAATCATAGAAGTCAACTCTTATTGAAAGCTTAGTCTGTG  
CCAAGCTCTATTCTAAAATGTATTGACCCTAAATACATTTAGATAATGTATACAT  
TATCTAAAATGTAATAAAAAAATAATAATTTTTGTTATTATTTCCATCCTATGGA  
TGAAGCTGCTGAAACCCAAGGTCCCCTACTGGGTGGTGGCAACACTATGAGCTG  
TTGGGAAAATATACTAAATCTATTATTAAGTTTAAAGTAAATAAATTGAATATTA  
TATAATTATATGAATTAATCAAATATAATTAAATATATTTTCAATAATAATATTA  
TTAGAAATAATTCAAGGAGAAAAAAATCCCCCAACCCAGAAAAACCTGAGG
```

substrings/kmers
of length 8



```
'TTTTTTTTT', 19  
'AAAAAAAT', 3  
'AAATAATT', 3  
'TAATTTTTT', 3  
'TATAATTA', 2  
'ATATAATT', 2  
'AATATTAT', 2  
'TATTATTA', 2  
'AAATATAA', 2  
'AAAATAAT', 2  
'AAAATGTA', 2  
'TAAAATGT', 2  
'CTAAAATG', 2  
'TCTAAAAT', 2  
'AATTGAAT', 2  
'TAATTGAA', 2  
'TAATTAAA', 2  
...
```

Identify what is

- unique
- frequent
- differentially frequent

...

gapped-kmer SVMs

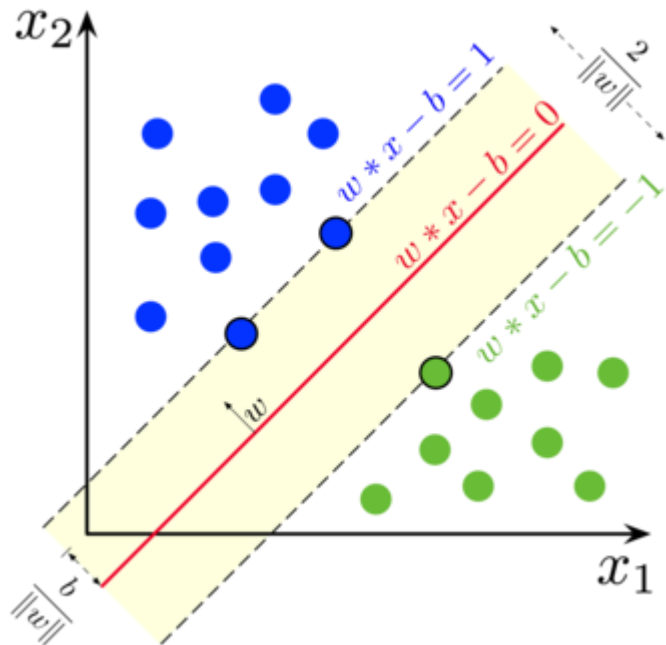
Gaps as „no information“ positions

GGAA TTGTAG

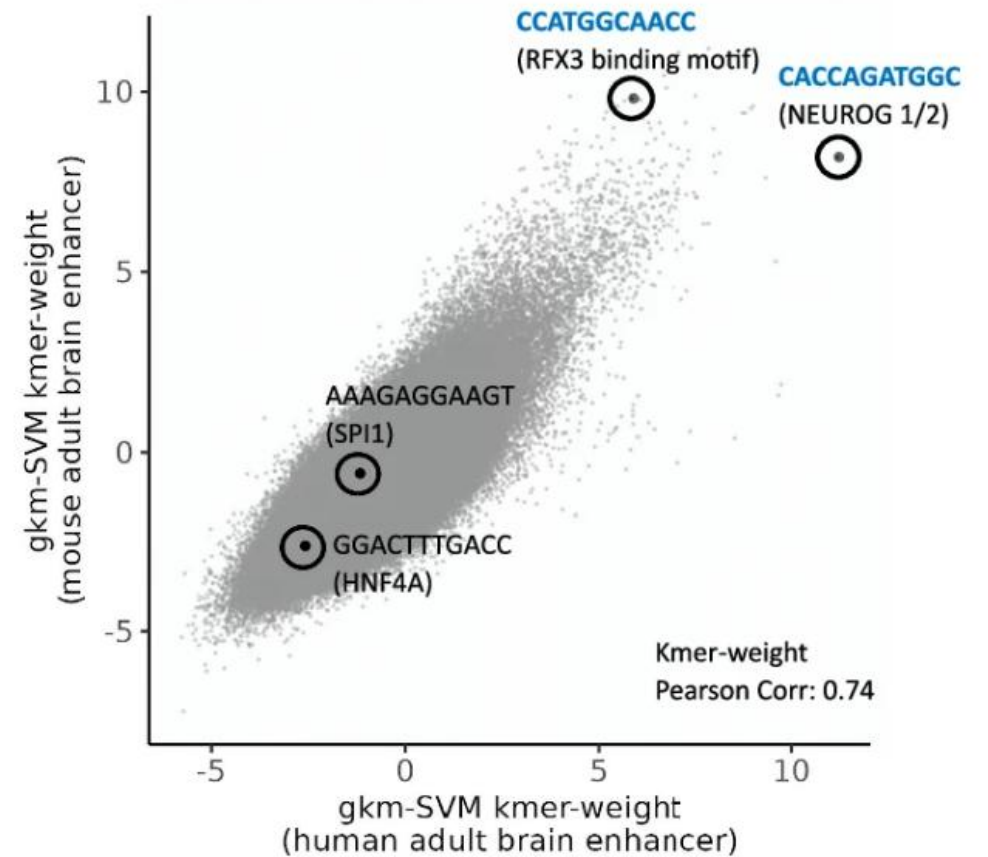
-G--TTGTAG

CGGT TTGTAG

-G--TTGTAG



Regulatory vocabularies
of gkm-SVM brain enhancer models



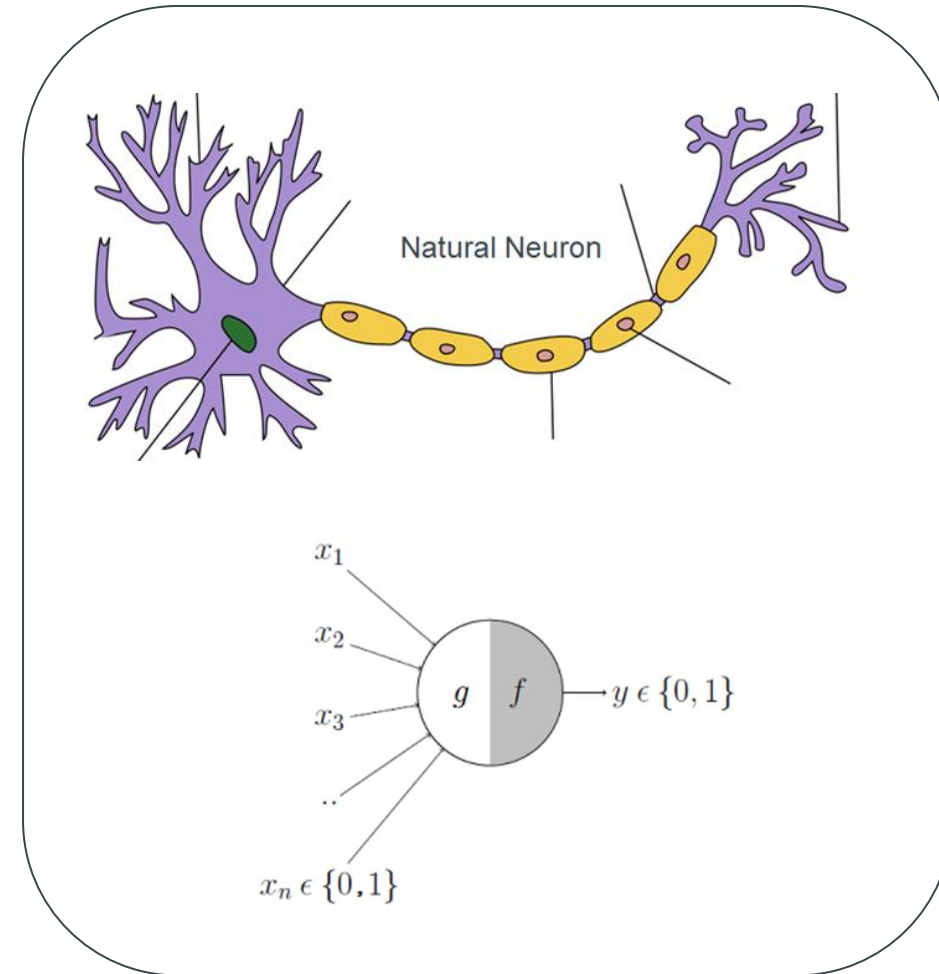
Neural networks (NNs)

Artificial
Neural
Networks

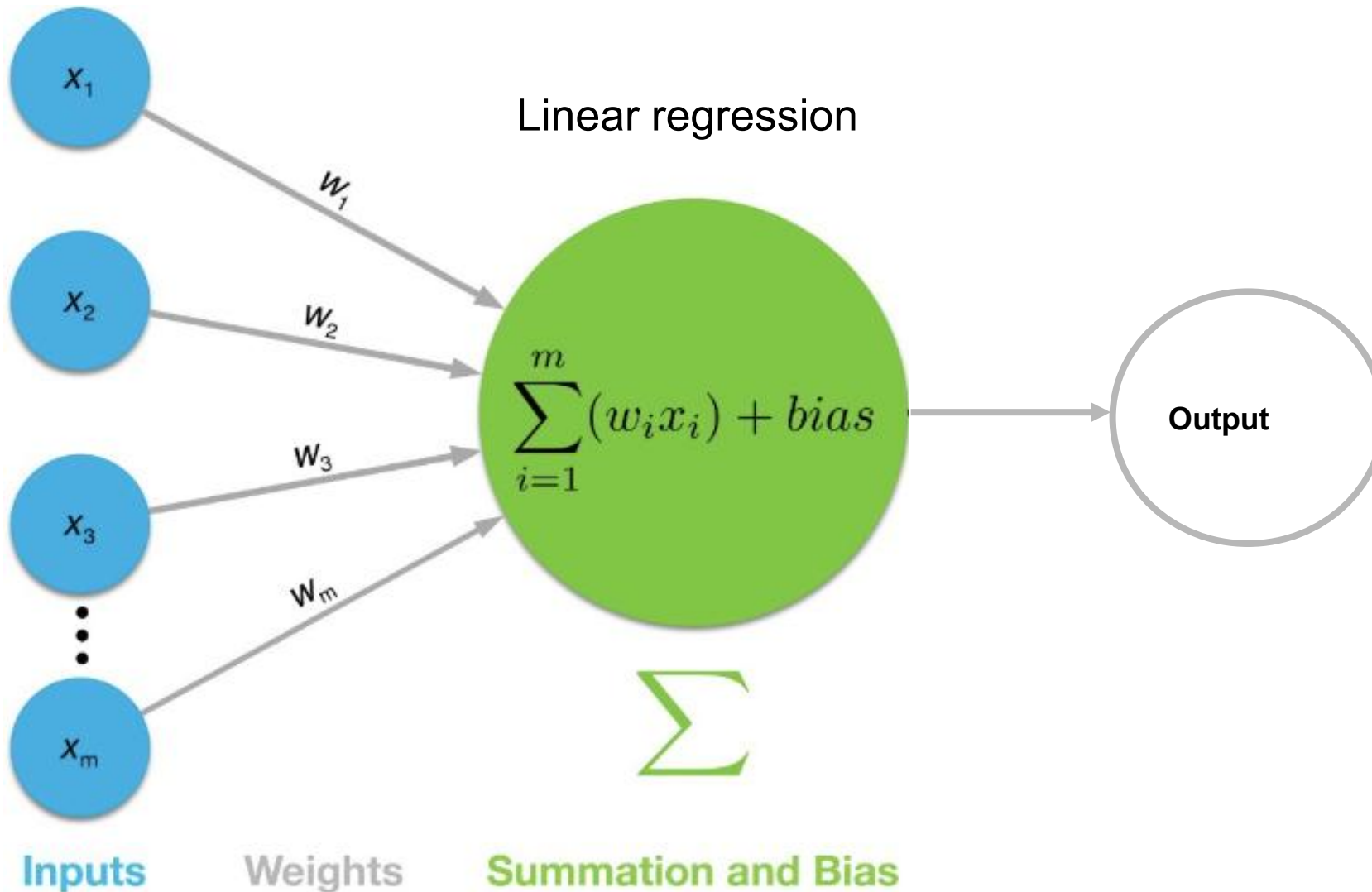
Born in the 1950s
Died in the 1970s

Reborn in 1980s
Died again in the 1990s

Reborn in the 2010s
Currently booming field



Neural networks (NNs)



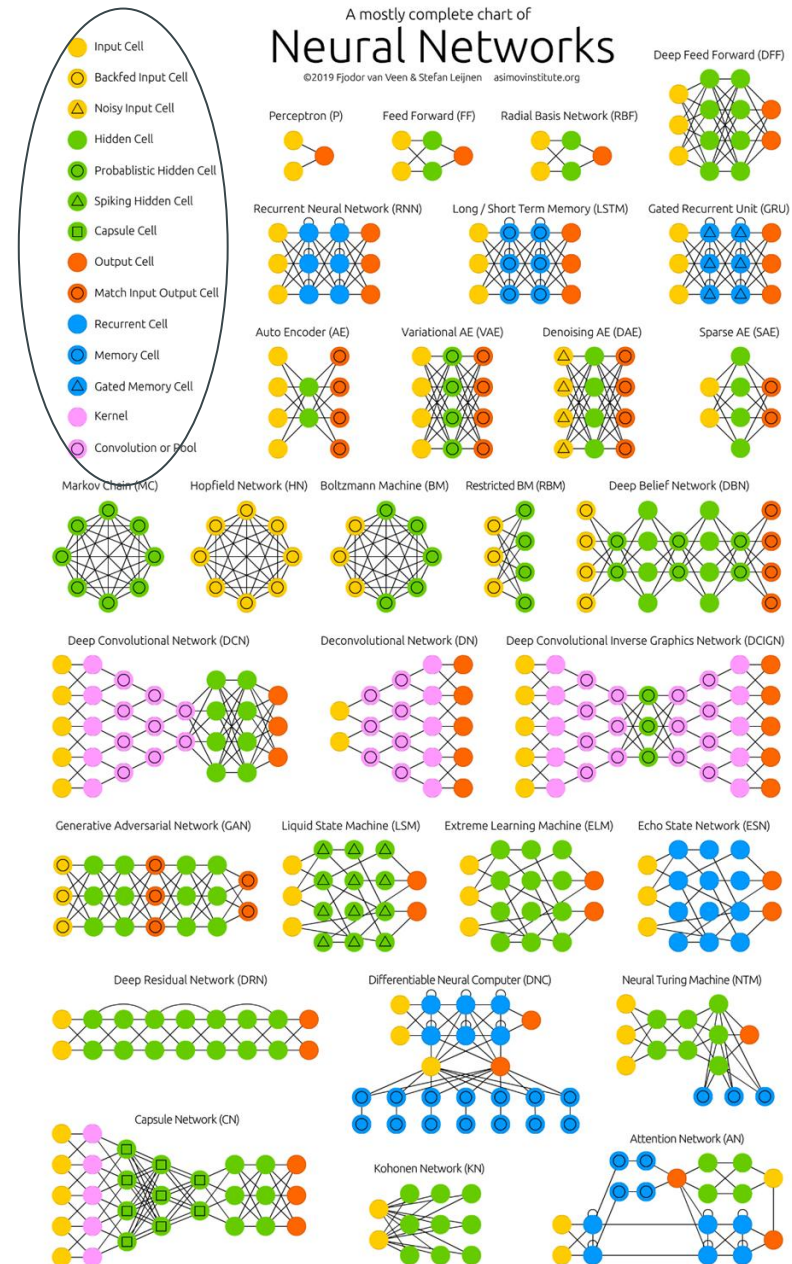
Deep Neural networks (NNs)

simple
building
blocks

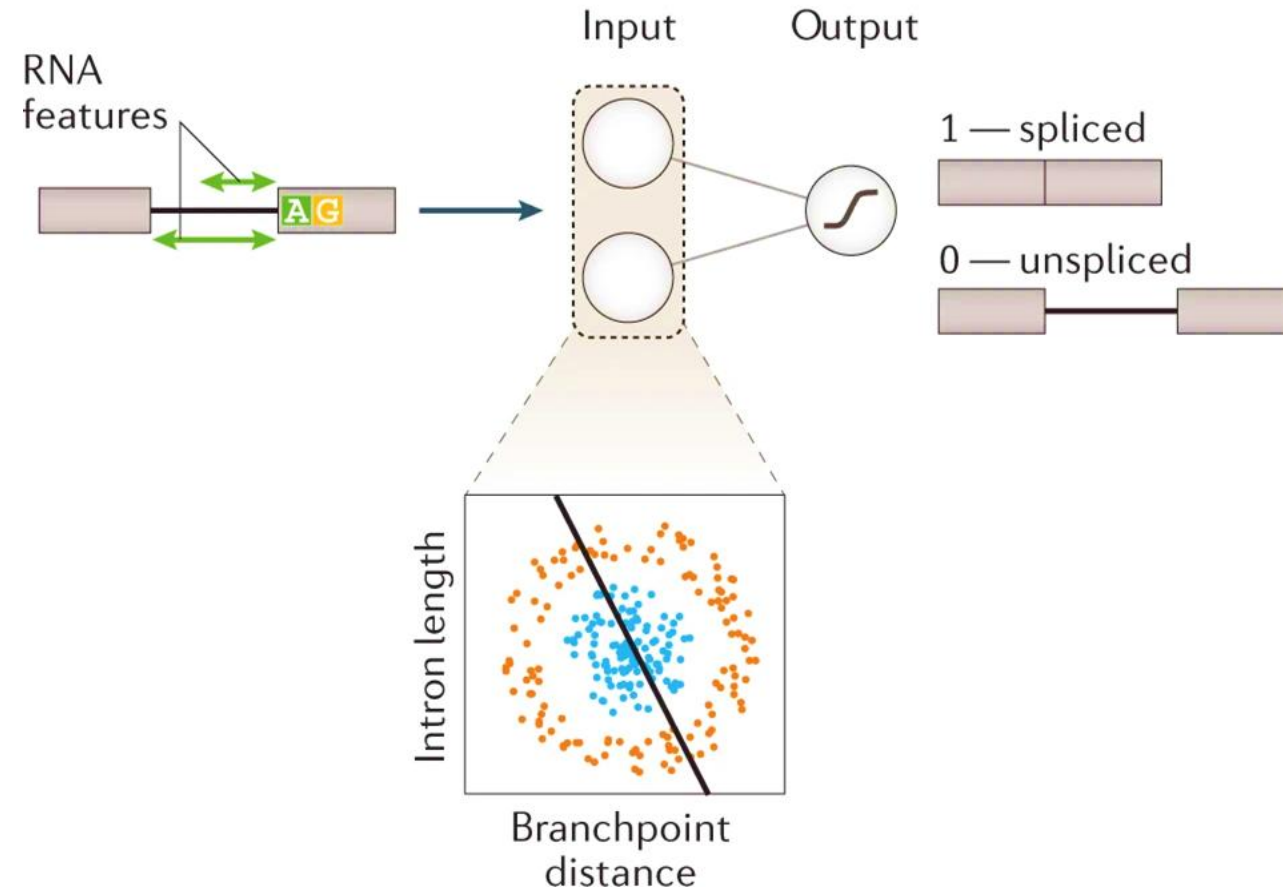
neurons

Deep Neural Networks

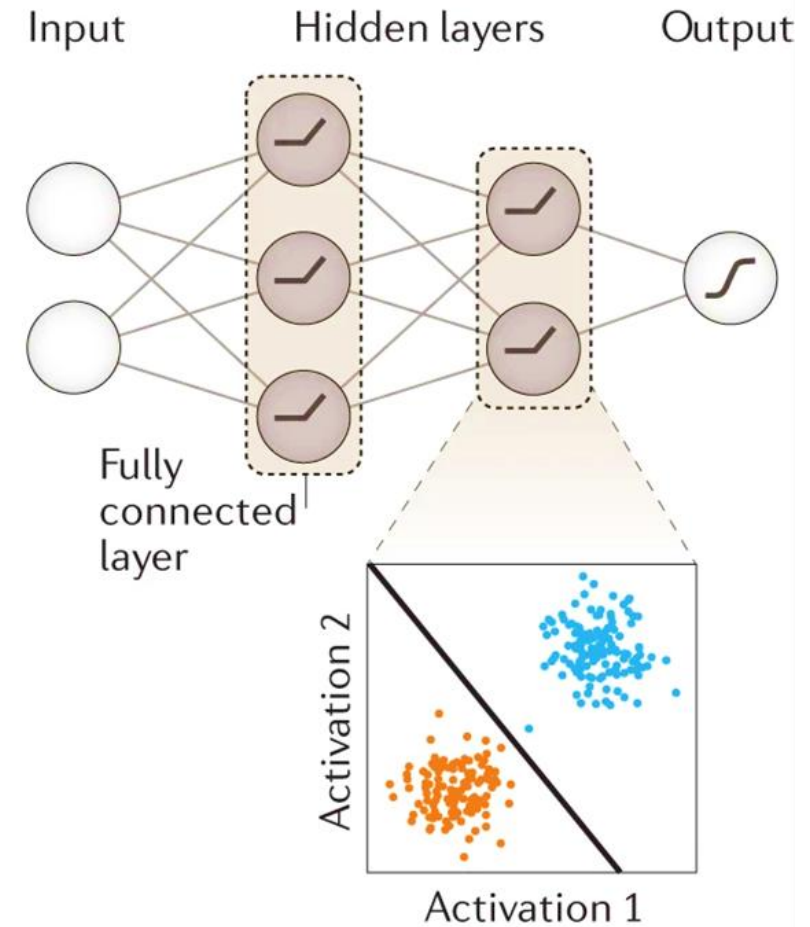
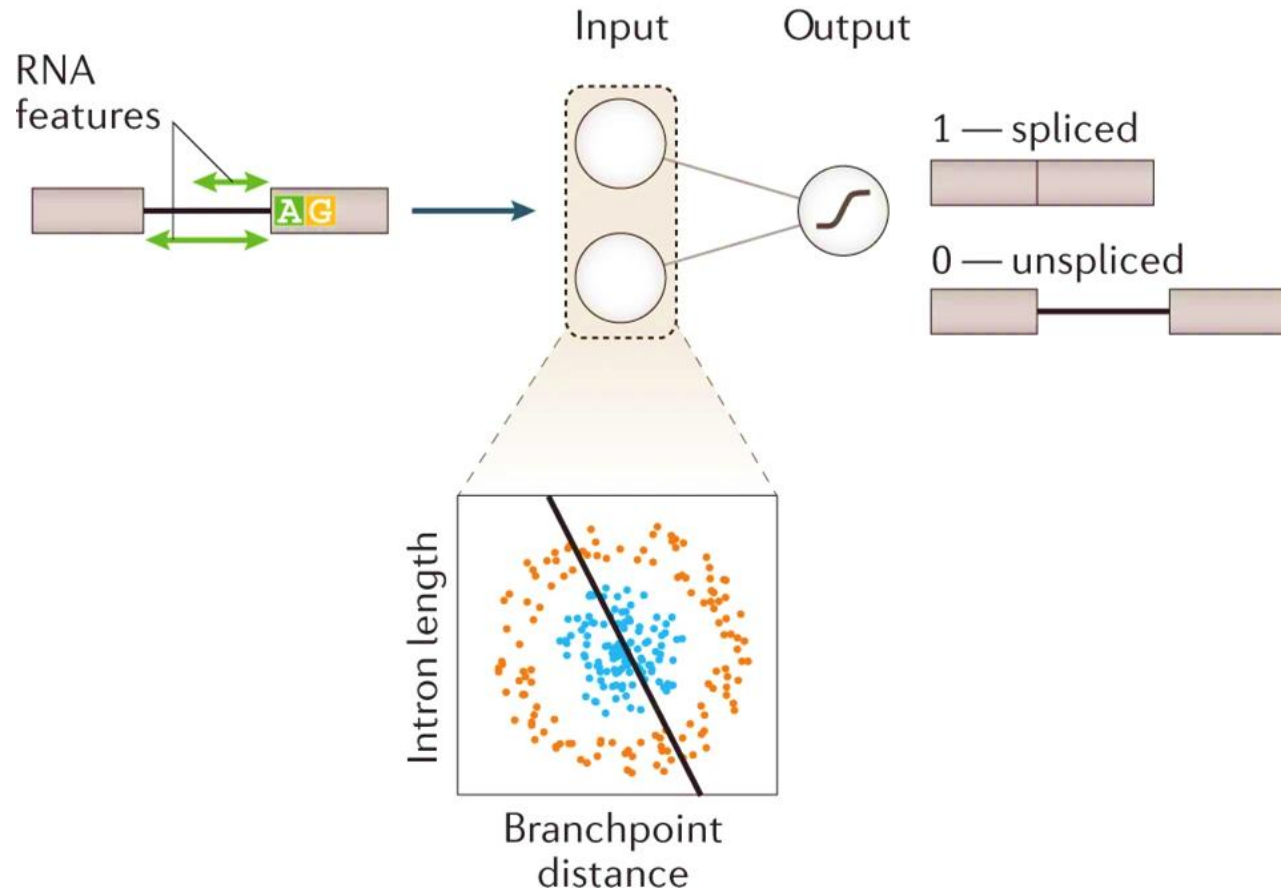
use **several layers** of
neurons
to model increasingly
complex
patterns in data.



Why neural networks? (NNs)



Enter Deep Neural Networks (DNNs)



Convolutional Neural networks (CNNs)

Looks for patterns in input (for e.g., an image), independently of their location

Matrix defines the pattern (filter)

Multiplying the image $\rightarrow$ number representing the 'fit'



*

-1	-2	-1
0	0	0
1	2	1



*

-1	0	1
-2	0	2
-1	0	1



1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

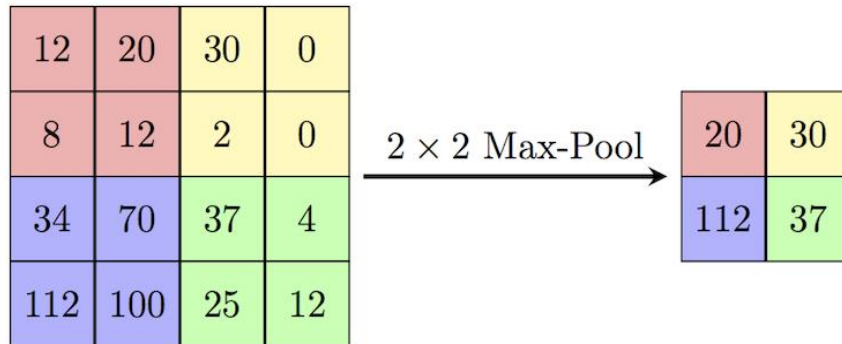
Image

4		

Convolved
Feature

Pooling layer

(Max.) - Pooling - taking the most significant areas

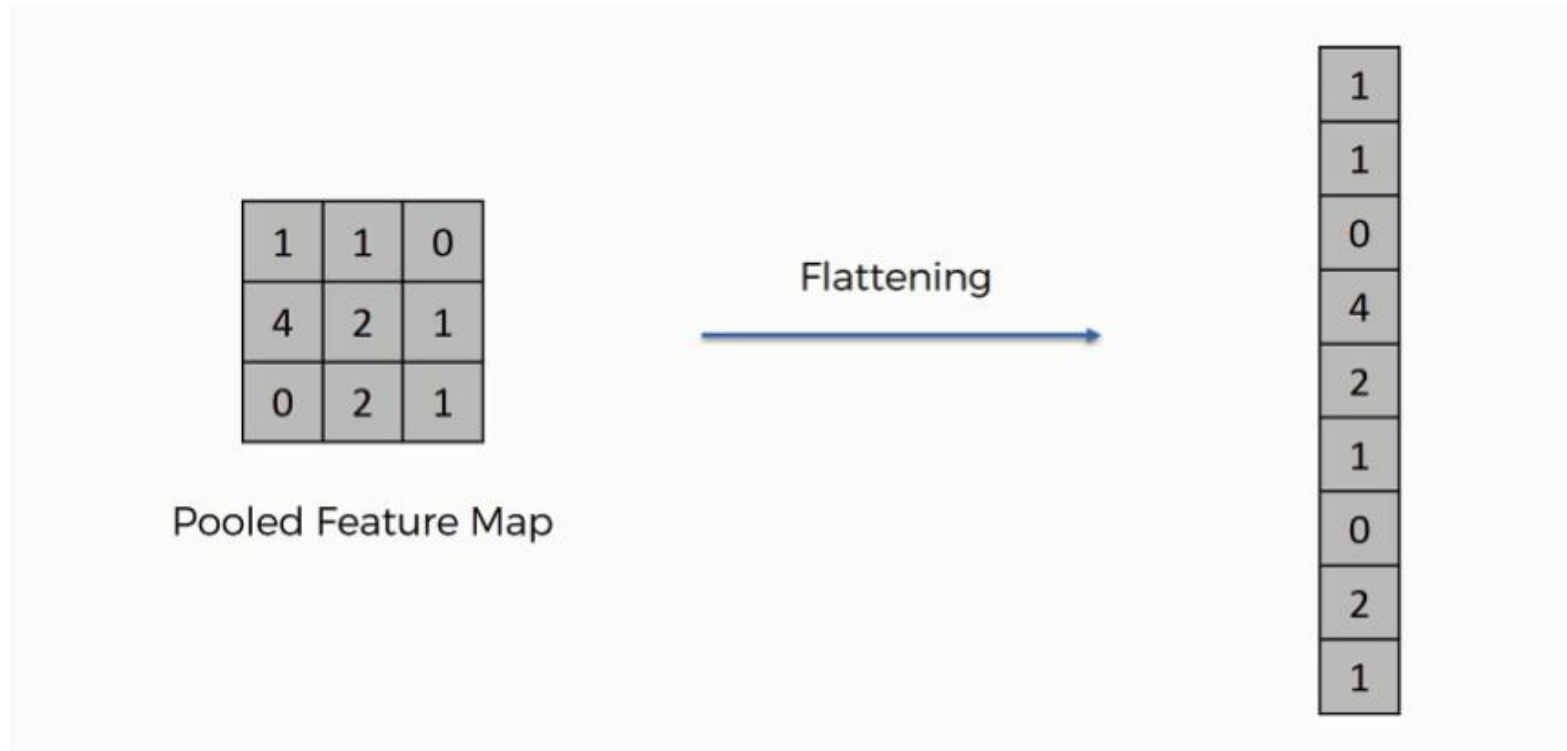


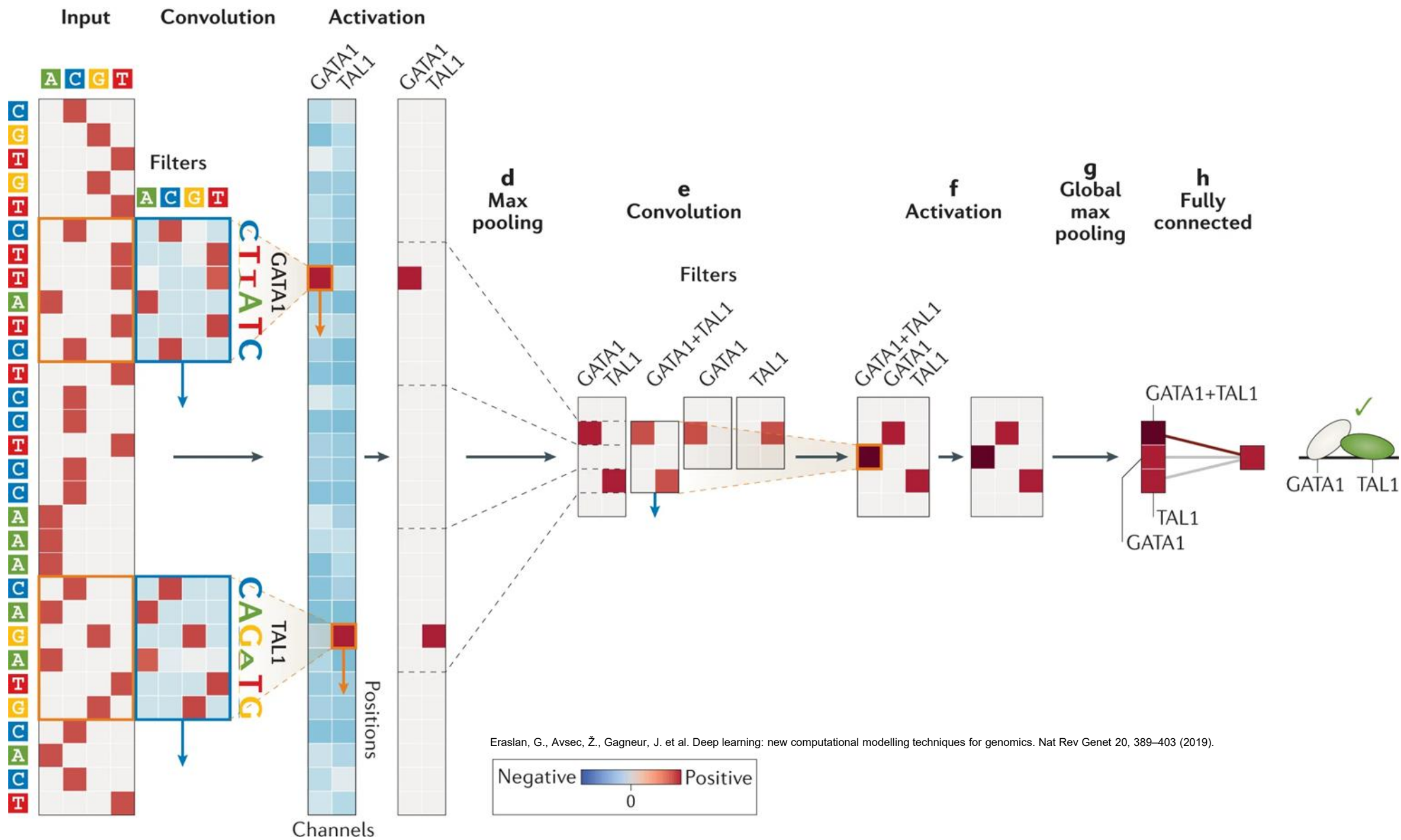
5	5	4	5
7	6	5	3
3	5	6	5
3	4	8	6

7	

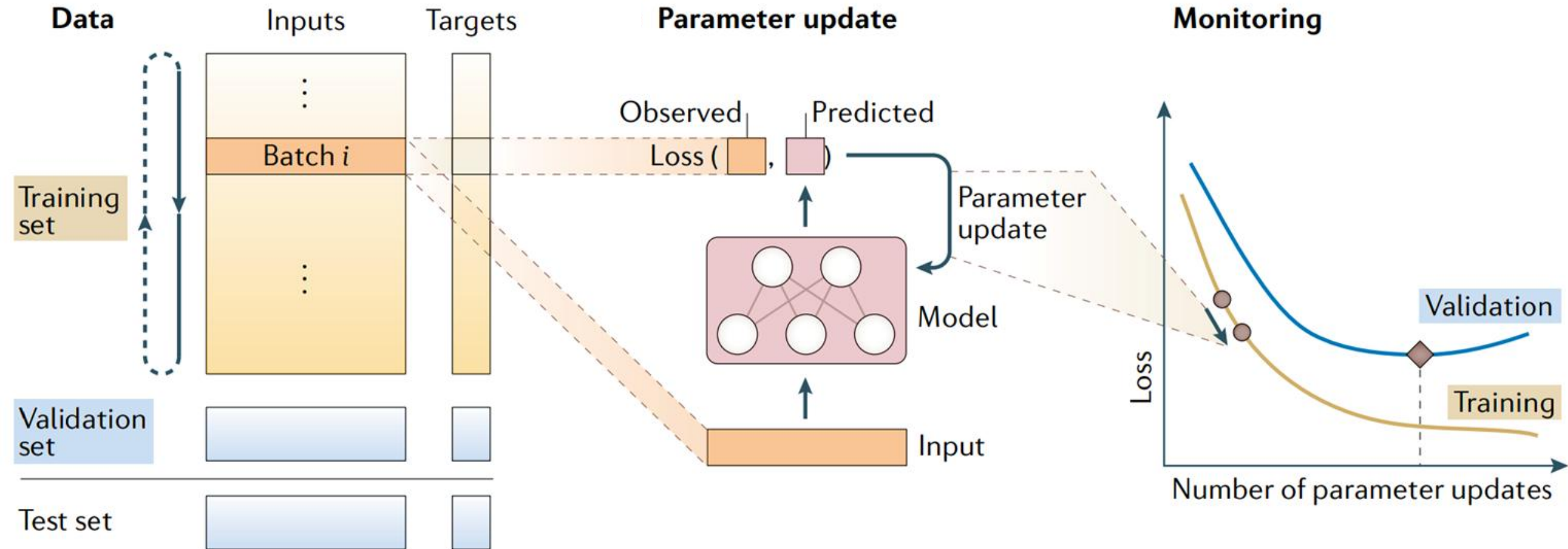
Flatten layer

Flatten - Spread the pixels and then take combinations of these

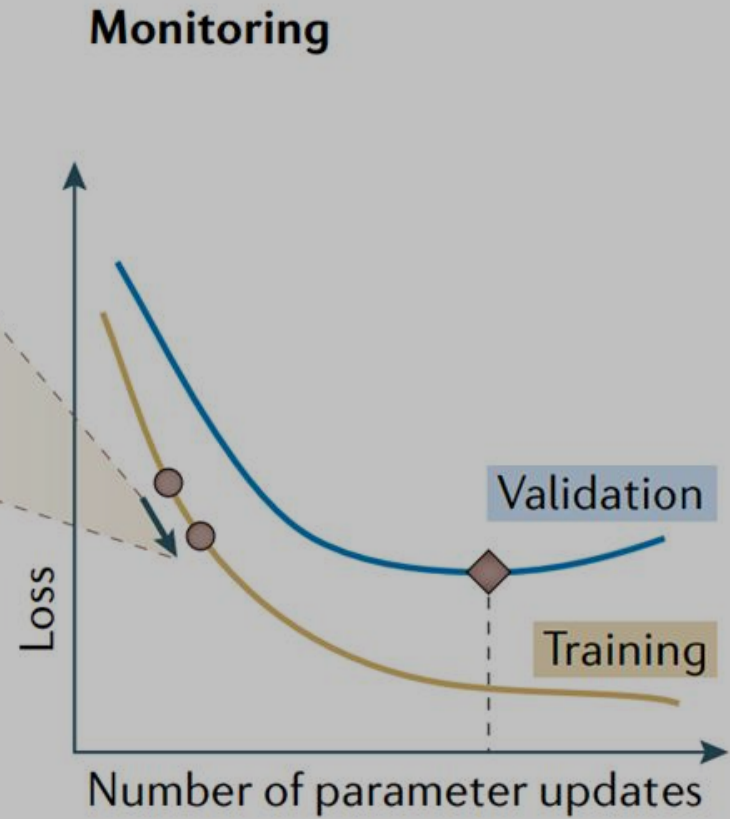
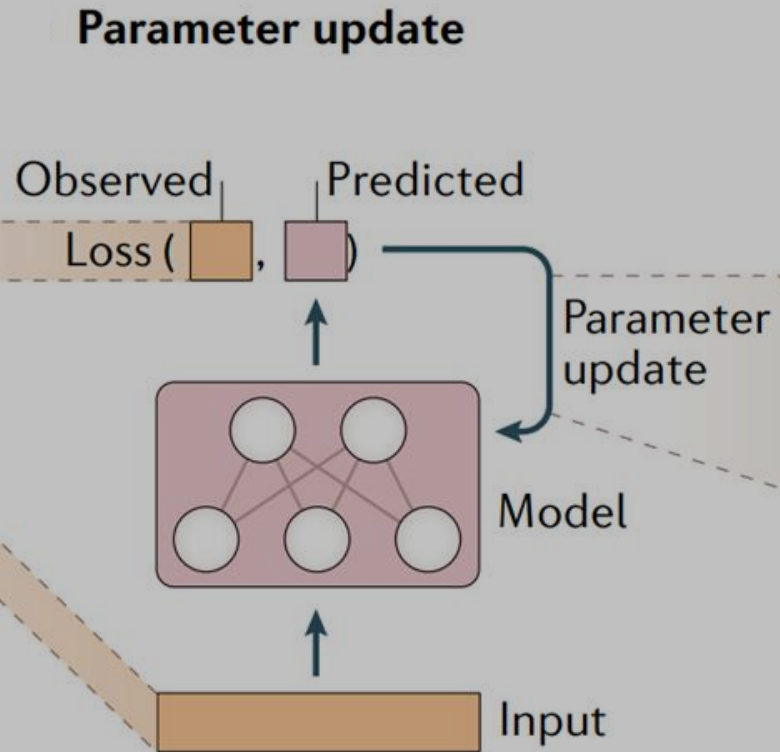
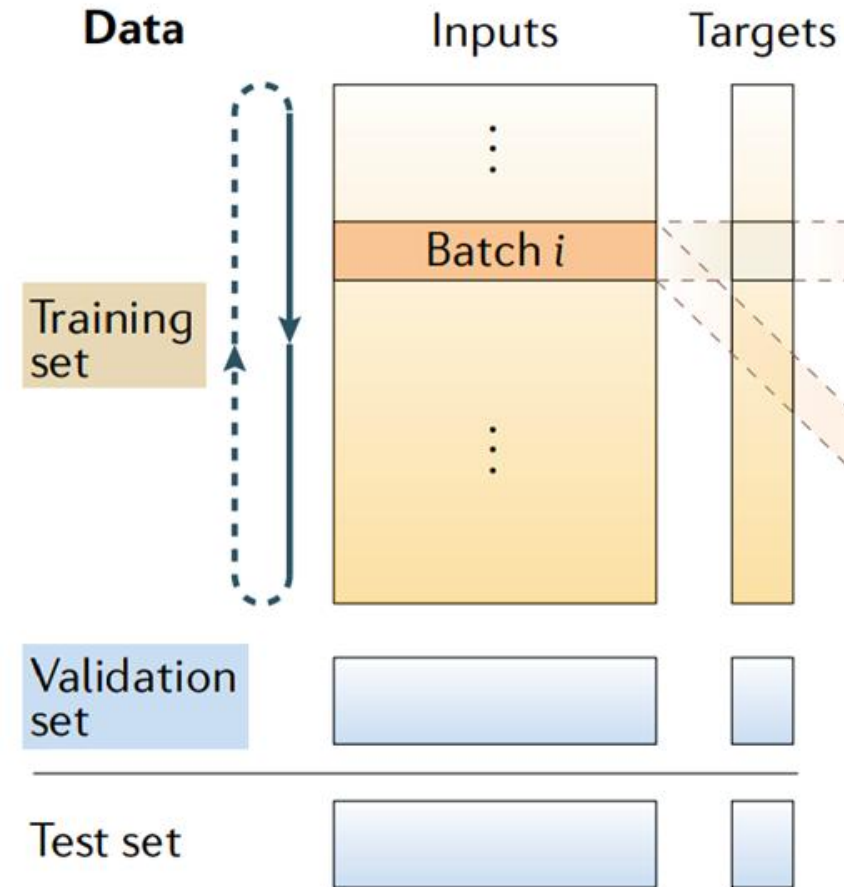


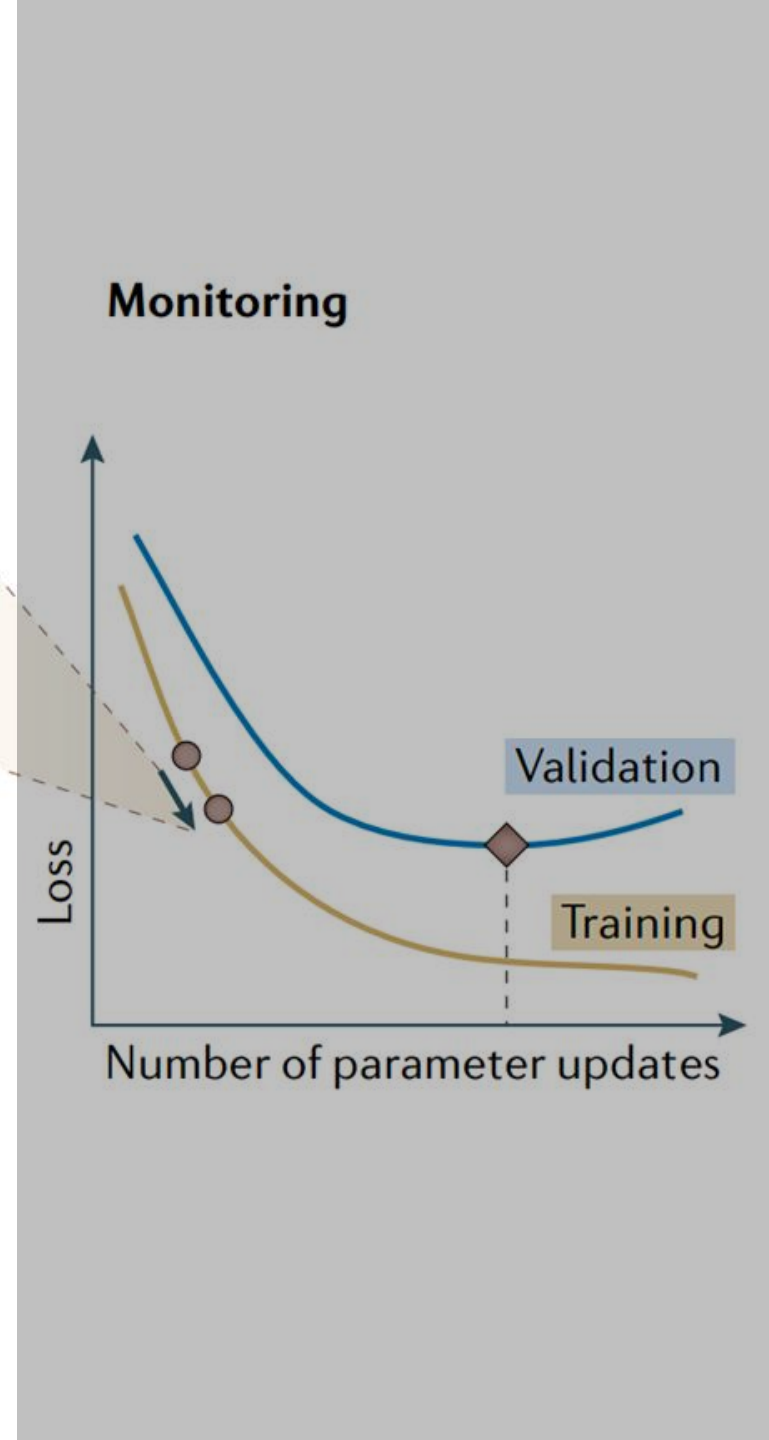
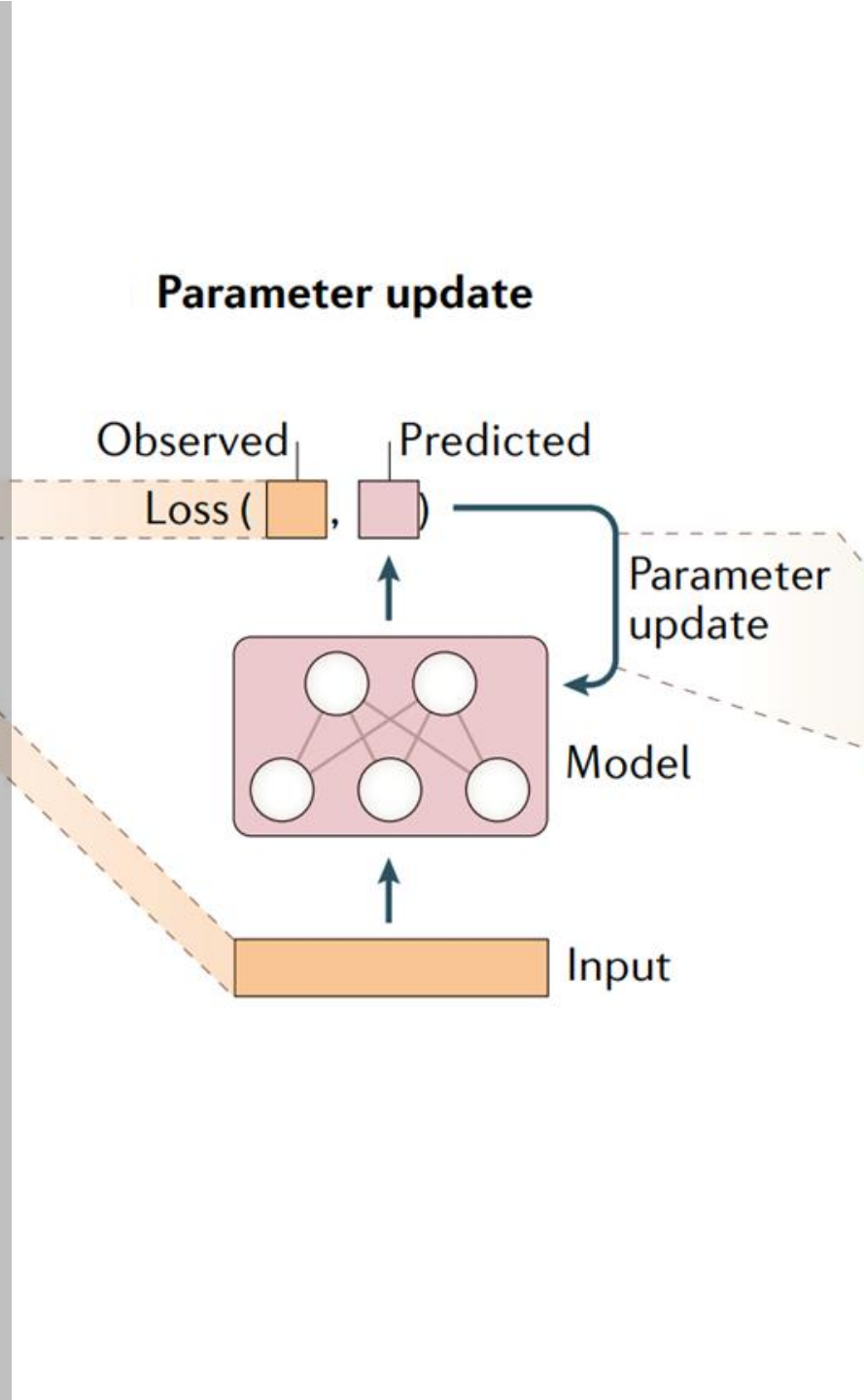
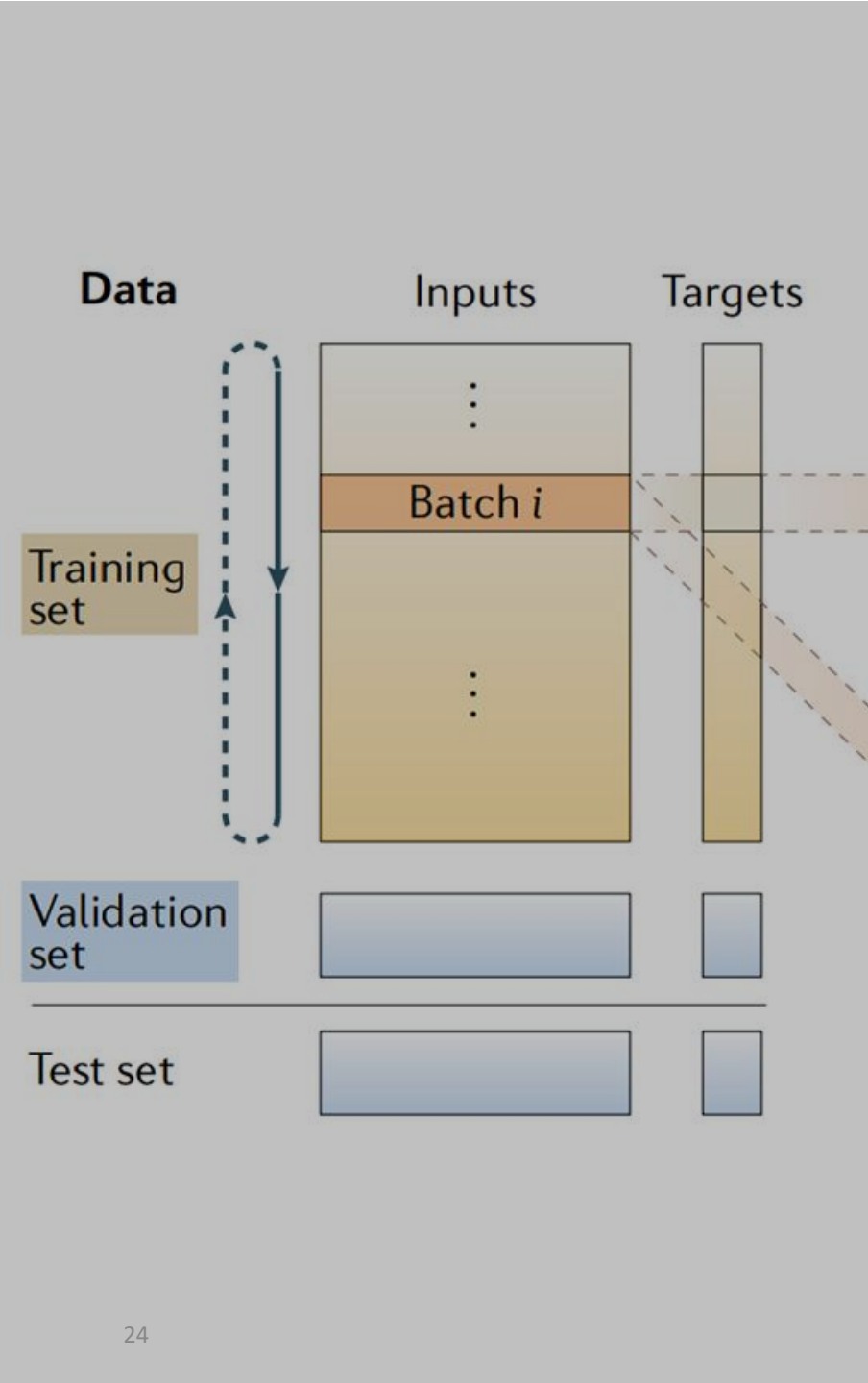


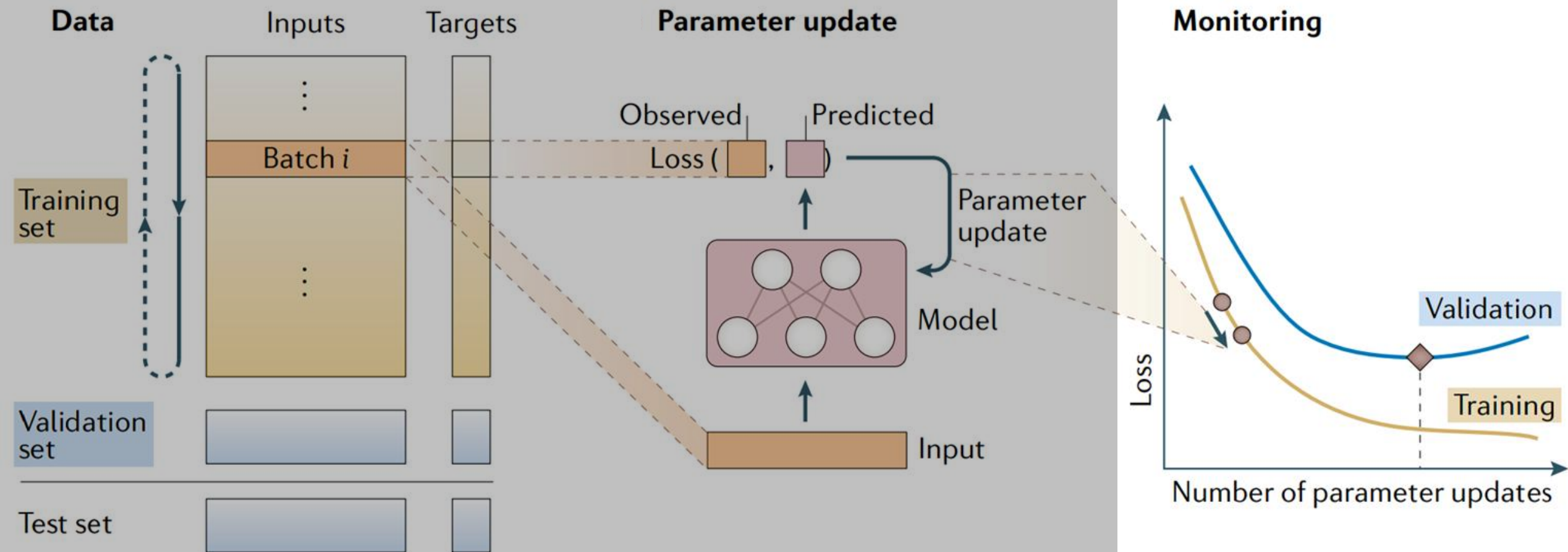
Model training



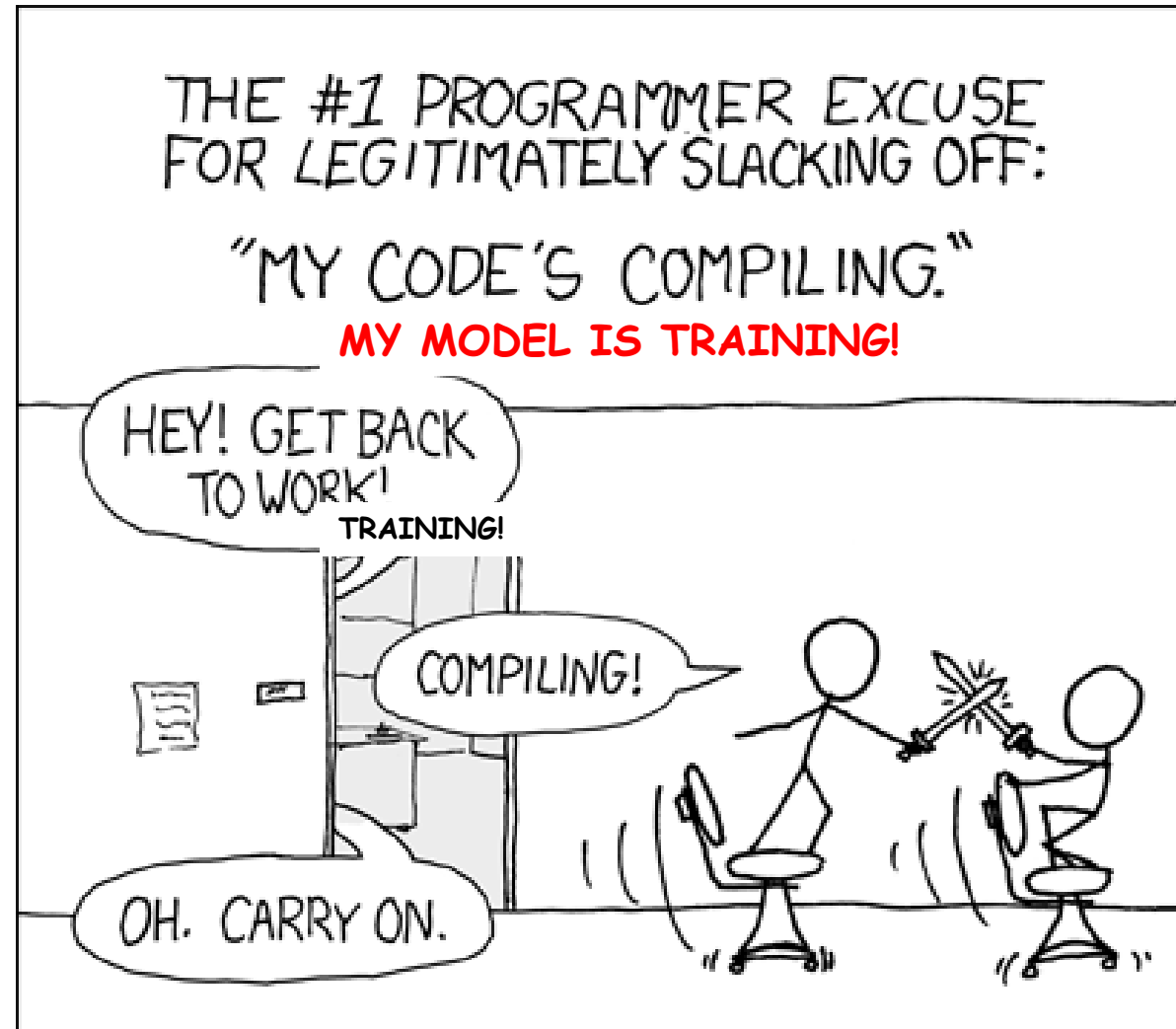
Model training





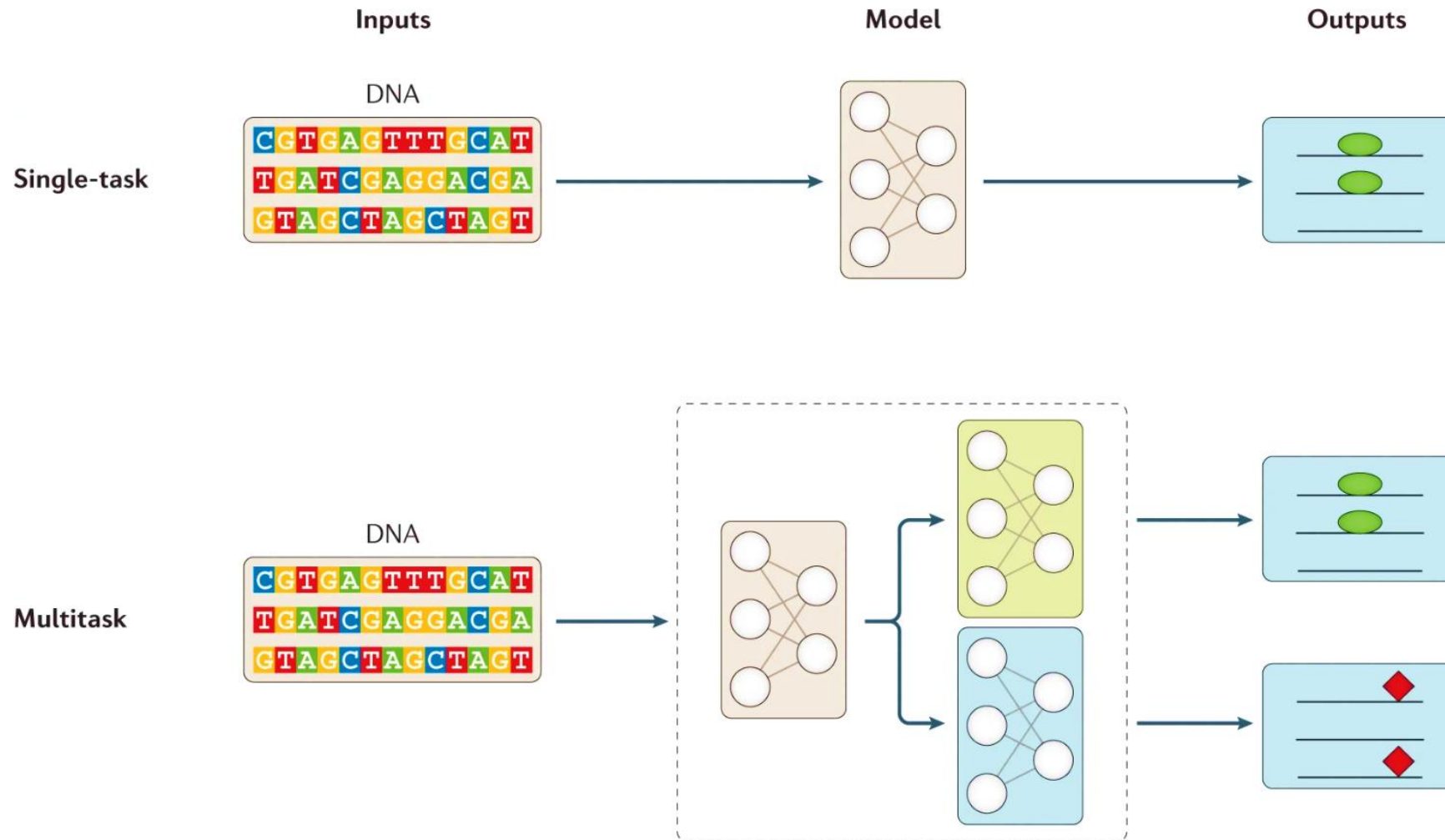


Hmm...

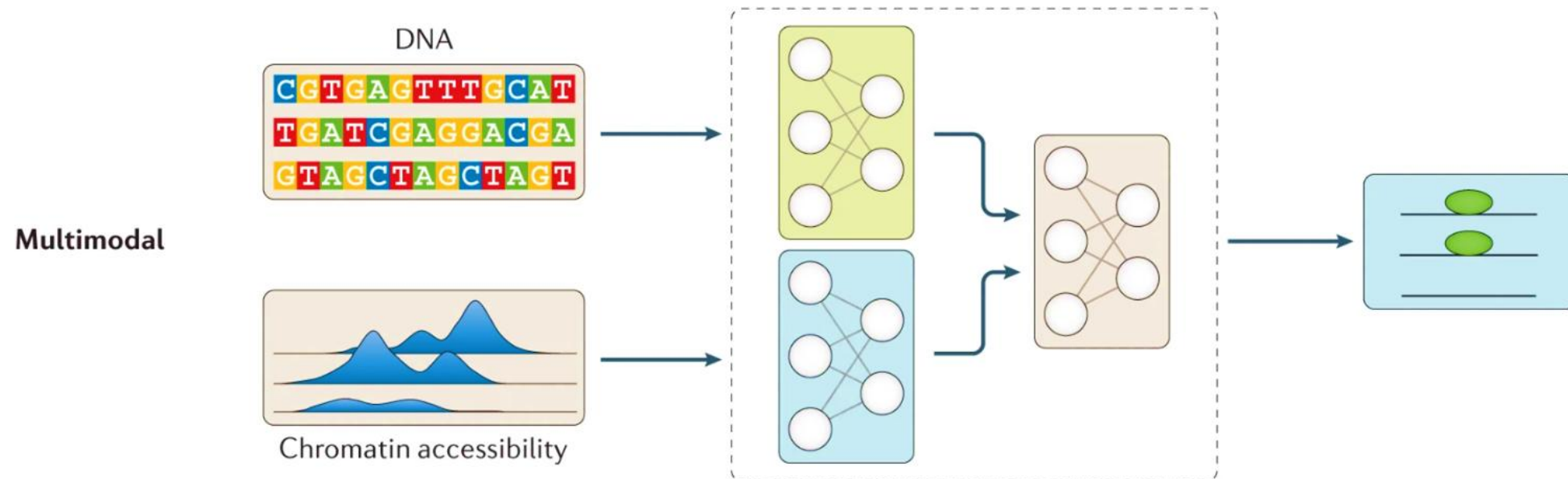


<https://xkcd.com/303/>

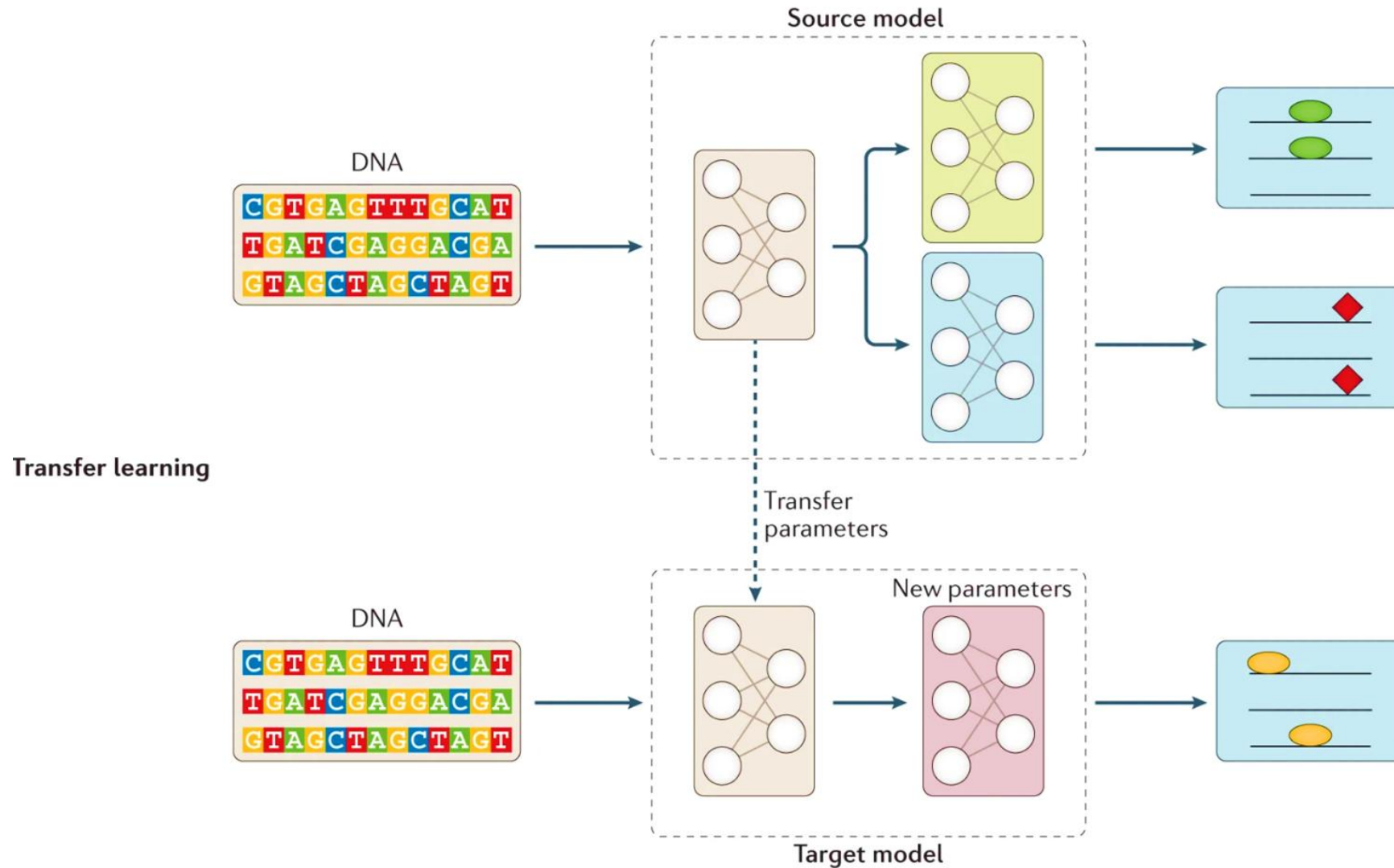
Model training – How to *stir the pile*?



Model training – How to *stir the pile*?



Model training – How to *stir the pile*?



Useful resources

Goodfellow, I., Bengio, Y. & Courville, A. *Deep Learning* (MIT Press, 2016). This textbook covers theoretical and practical aspects of deep learning with introductory sections on linear algebra and machine learning.

Eraslan, G., Avsec, Ž., Gagneur, J. *et al.* Deep learning: new computational modelling techniques for genomics. *Nat Rev Genet* 20, 389–403 (2019).

<https://huggingface.co/>

The AI community building the future.

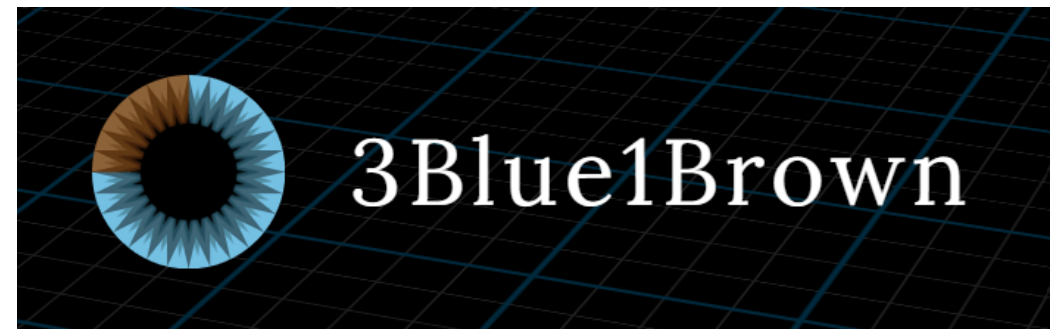


Kipoi

<https://kipoi.org/>

Model Zoo for genomics

<https://www.3blue1brown.com/>



Useful resources (2)

<https://www.kaggle.com/>



Critical Assessment of Genome Interpretation

Challenges

Meetings

Publications

Organization

Contact Us

<https://genomeinterpretation.org/>

Challenges

<https://paperswithcode.com/>



SageBionetworks

About ▾

How We Partner ▾

Tools & Resources ▾

Careers ▾

News ▾

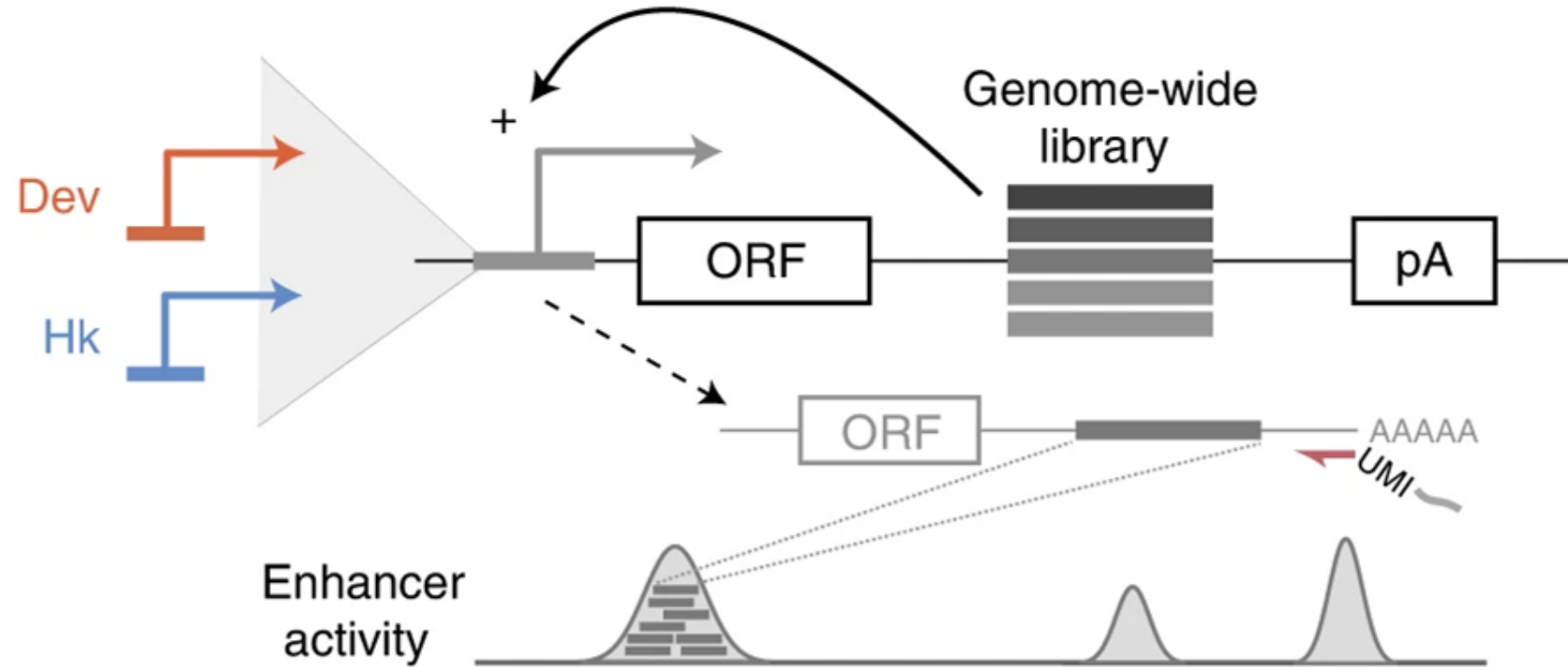
Donate ▾

https://sagebionetworks.org/tools_resources/challenges/

Challenge Platform

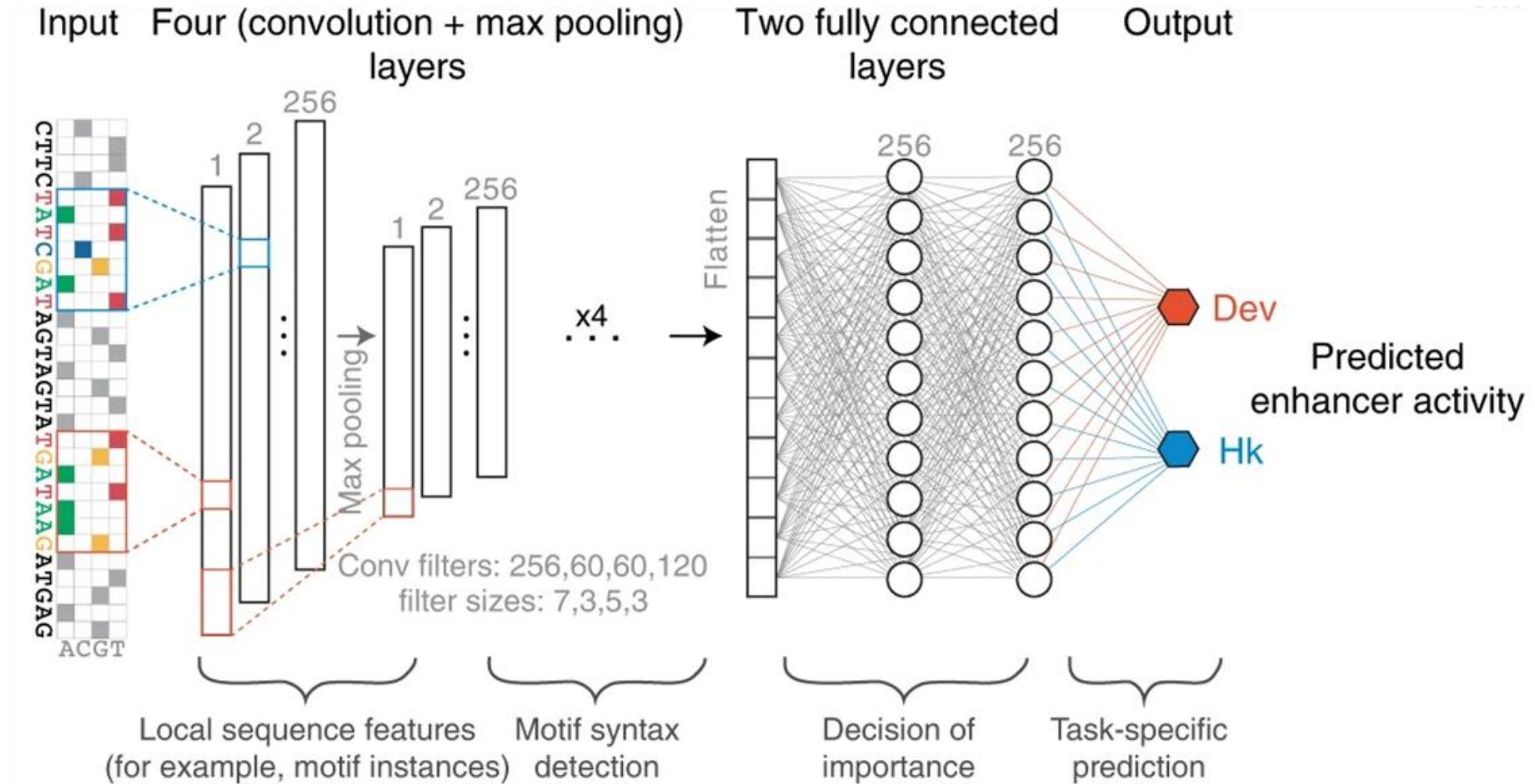


Example: DeepSTARR



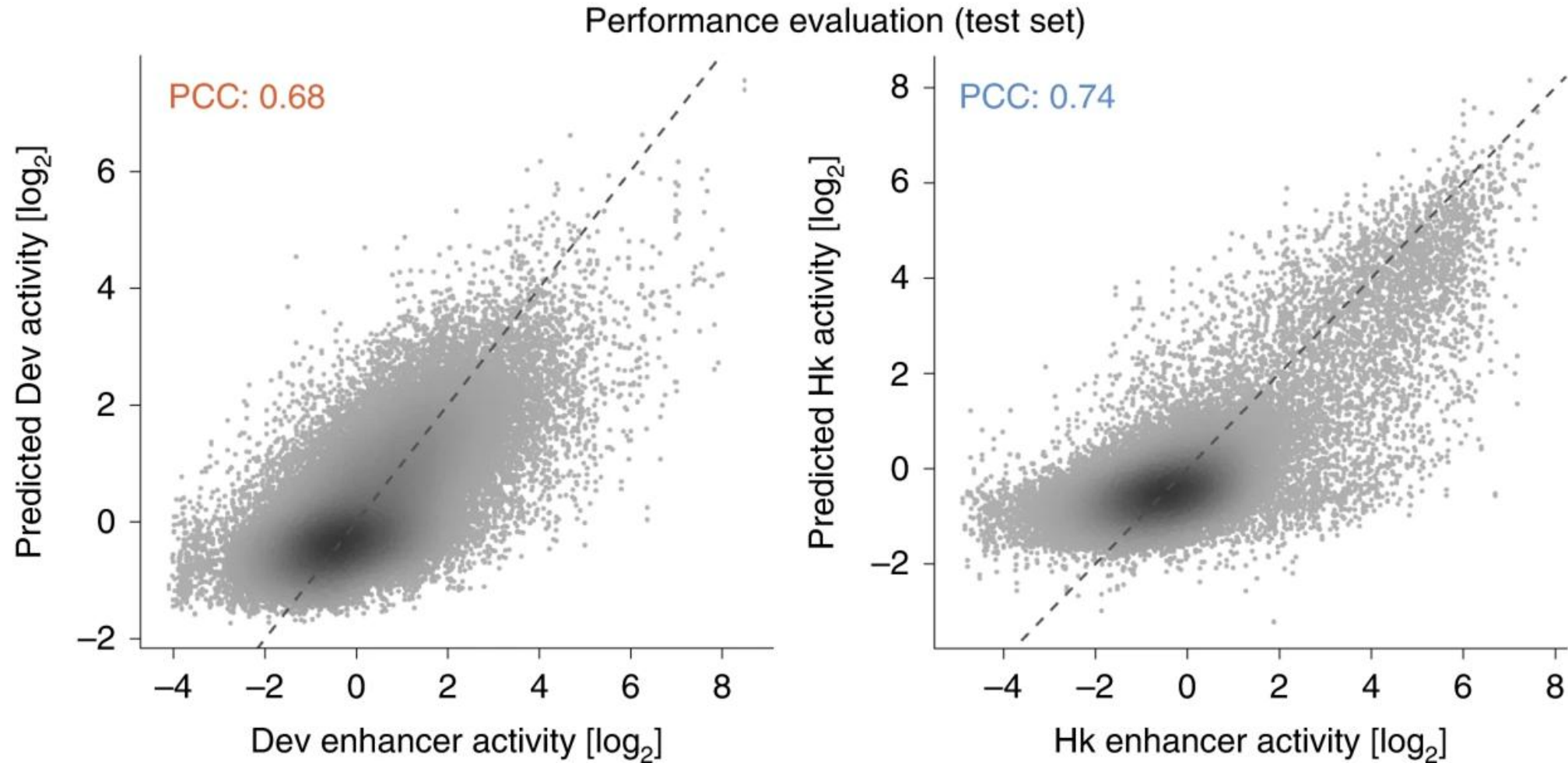
de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

Example: DeepSTARR



de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

Example: DeepSTARR



de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

Acknowledgements

- Pyaree M. Dash (BIH – Berlin, Germany)
- Thorben Maaß (UKSH – Lübeck, Germany)

CONTACT

Arjun Devadas arjun.devadas@uni-luebeck.de
Martin Kircher martin.kircher@uni-luebeck.de
Max Schubach max.schubach@bih-charite.de



kircherlab.github.io



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein

BIH Berlin Institute
of Health
@Charité

Hands-On sessions: Model training



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and iGVF Resource)

Arjun Devadas

Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

Hands-on: Training a sequence-based model



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Go to: 05_sequence_models/01_training_sequence_model.ipynb

Klick: Open in Colab

Discussion of pre-training and fine-tuning approaches



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and IGVF Resource)

Arjun Devadas, Kilian Salomon, Max Schubach, Martin Kircher
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK

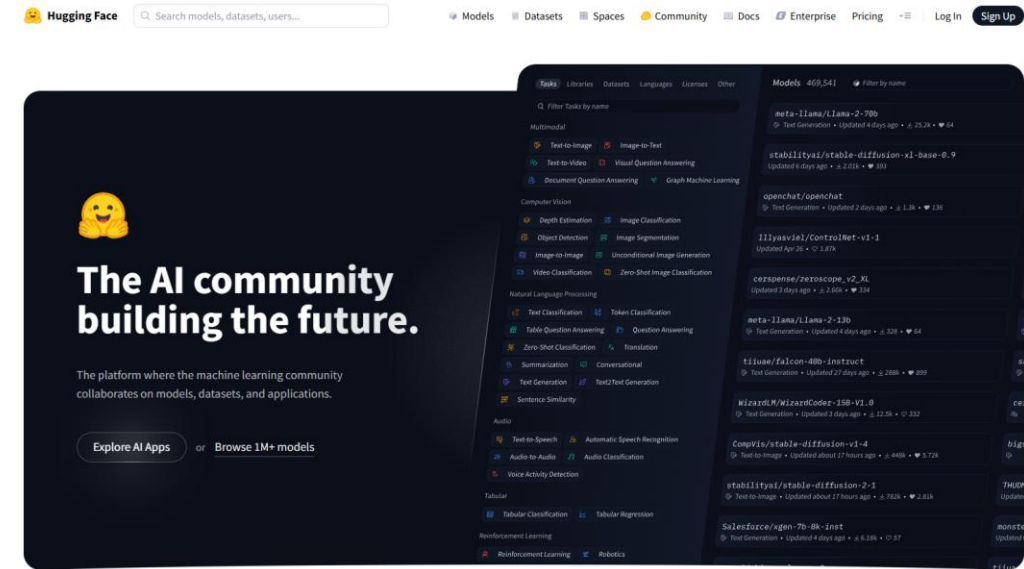


UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



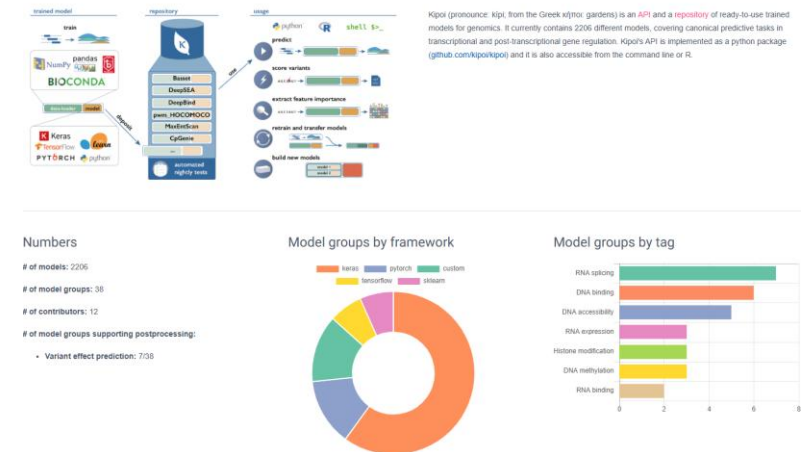
Benefiting from available resources

- **Accelerated Development:** Leverage pre-trained models to jump-start projects, reducing prototyping and training time.
- **Resource Efficiency, Flexibility and Adaptability:** Utilize models that have already been trained on massive datasets, saving on computational costs and data collection efforts.
- **Robust Feature Extraction:** Benefit from learned representations that capture complex patterns, making fine-tuning more effective for specific tasks.
- **Standardized Benchmarks:** Use well-established models as baselines to ensure reproducibility and make comparative evaluations more straightforward.



<https://huggingface.co/>

Kipoi: Model zoo for genomics

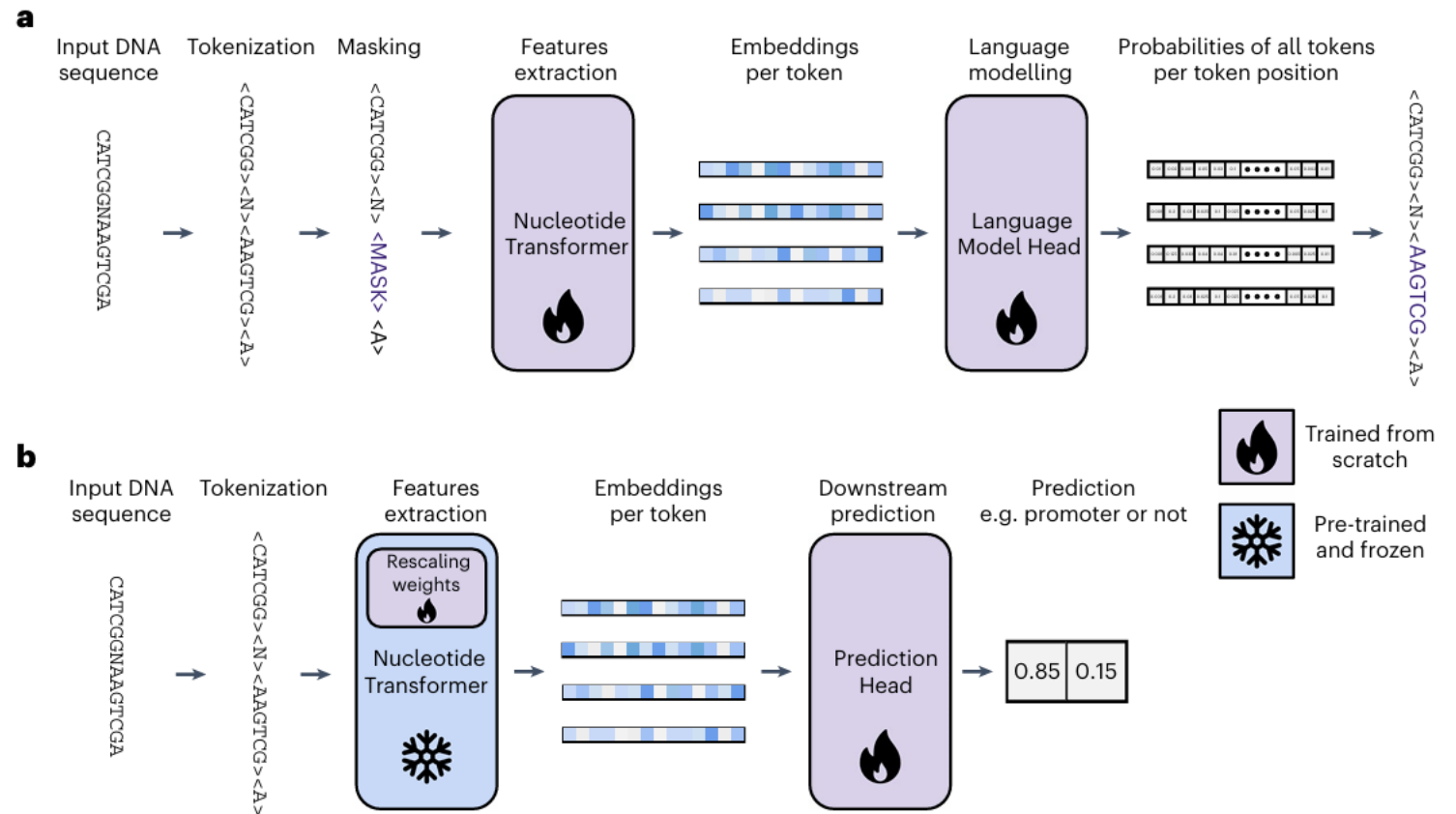


<https://kipoi.org/>

Foundation models like DNABERT, HyenaDNA, NT or Enformer

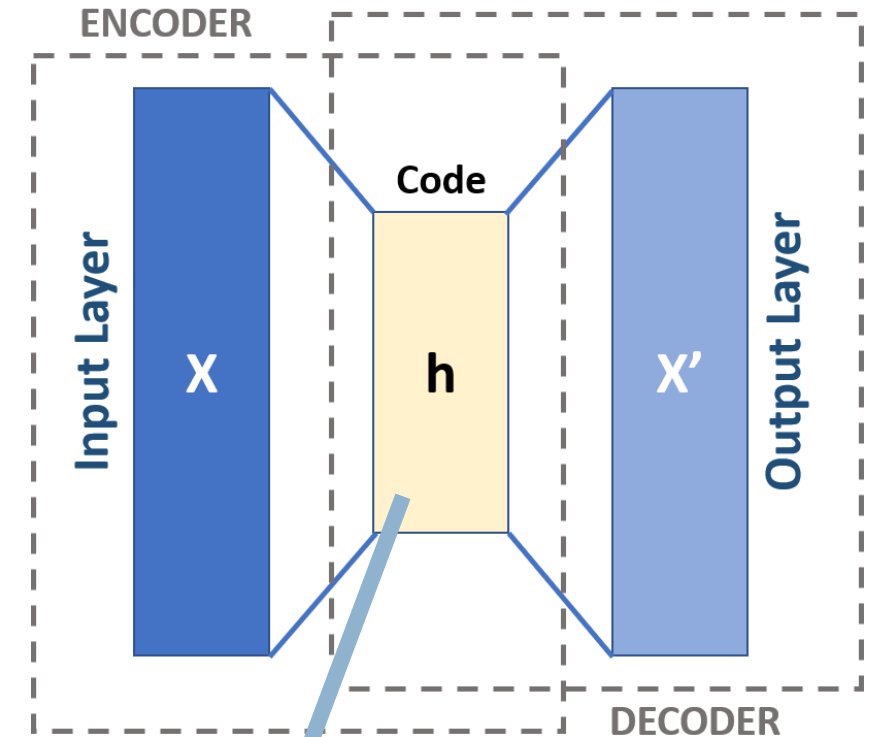
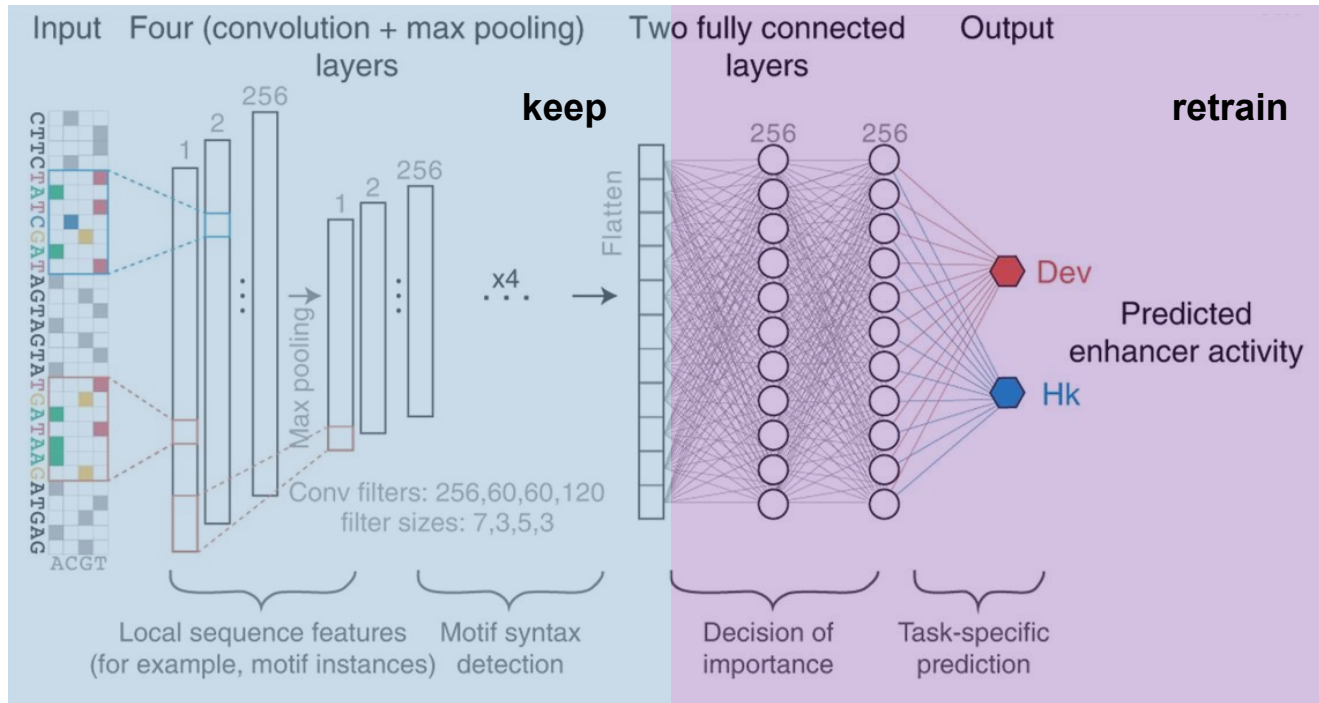
Foundation models are pre-trained on massive datasets, converting raw nucleotide sequences into rich, context-aware representations

Could refer specifically to language models (e.g. DNABERT, HyenaDNA, NT) but or could use all kinds of models.



Using pre-trained layers, data transformation from autoencoders, ...

DeepSTARR



https://de.wikipedia.org/wiki/Autoencoder#/media/Datei:Autoencoder_schema.png

use latent variables as input to
(classical) machine learning models

What is your experience or other insight?



Feel free to reach out!

kircherlab.github.io

CONTACT

Regulatory Genomics & Computational Genome Biology

Arjun Devadas arjun.devadas@uni-luebeck.de

Kilian Salomon kilian.salomon@bih-charite.de

Martin Kircher martin.kircher@uni-luebeck.de

Max Schubach max.schubach@bih-charite.de



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein

BIH Berlin Institute
of Health
@Charité

Hands-On sessions: Model interpretation



ISMB-ECCB 2025: Tutorial IP2 (MPRAs and iGVF Resource)

Max Schubach

Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025



UNIVERSITÄT ZU LÜBECK

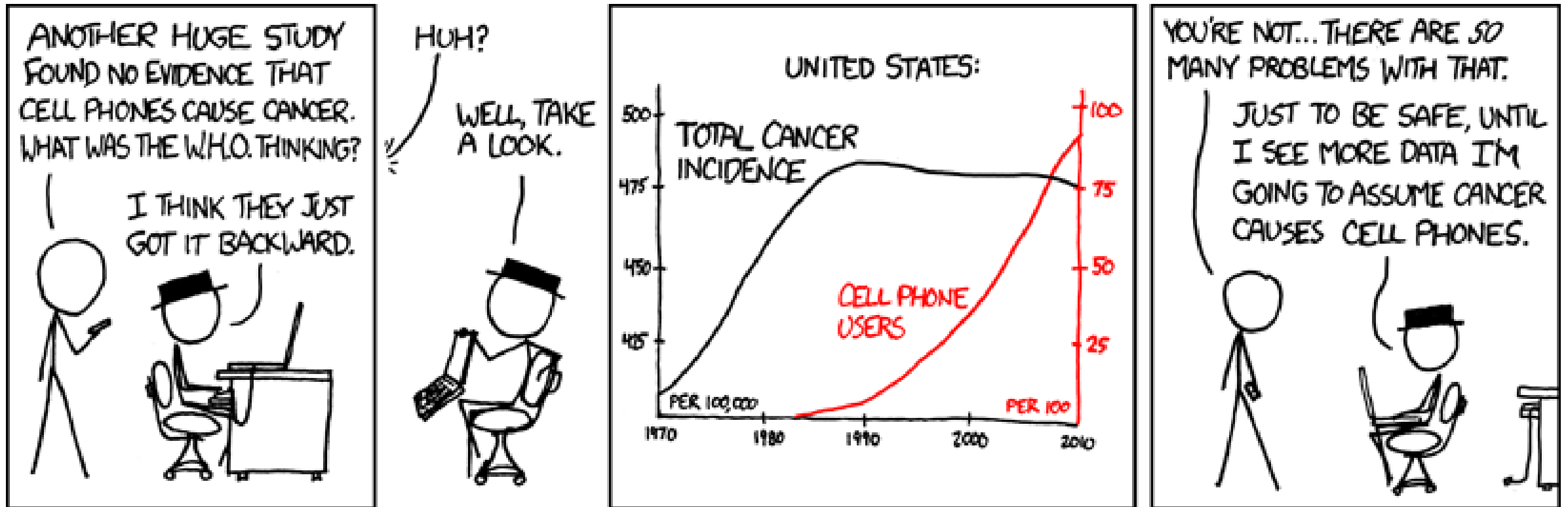


UNIVERSITÄTSKLINIKUM
Schleswig-Holstein



Berlin Institute
of Health
@Charité

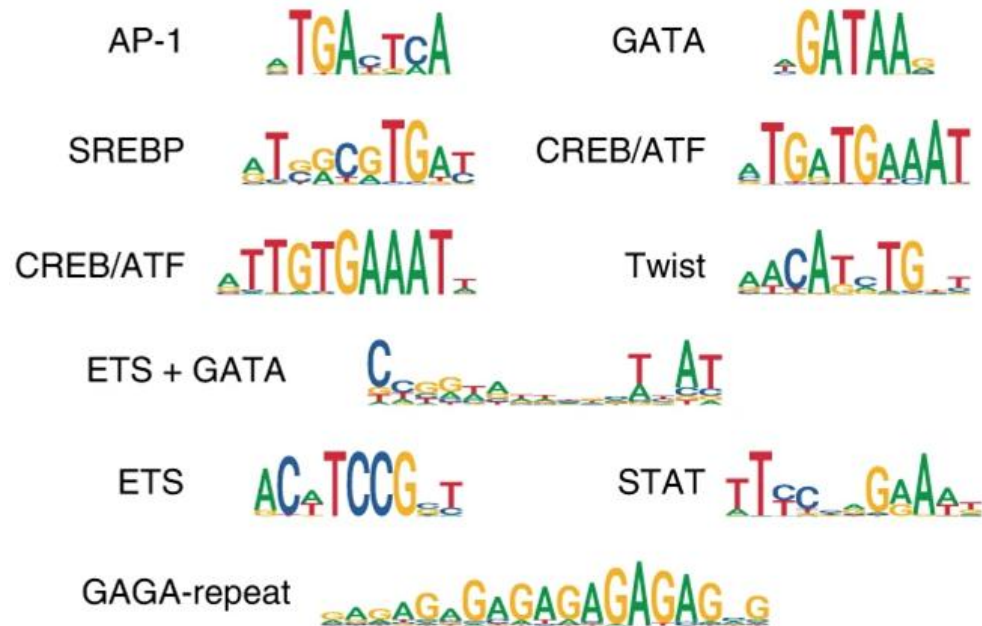
Model interpretation



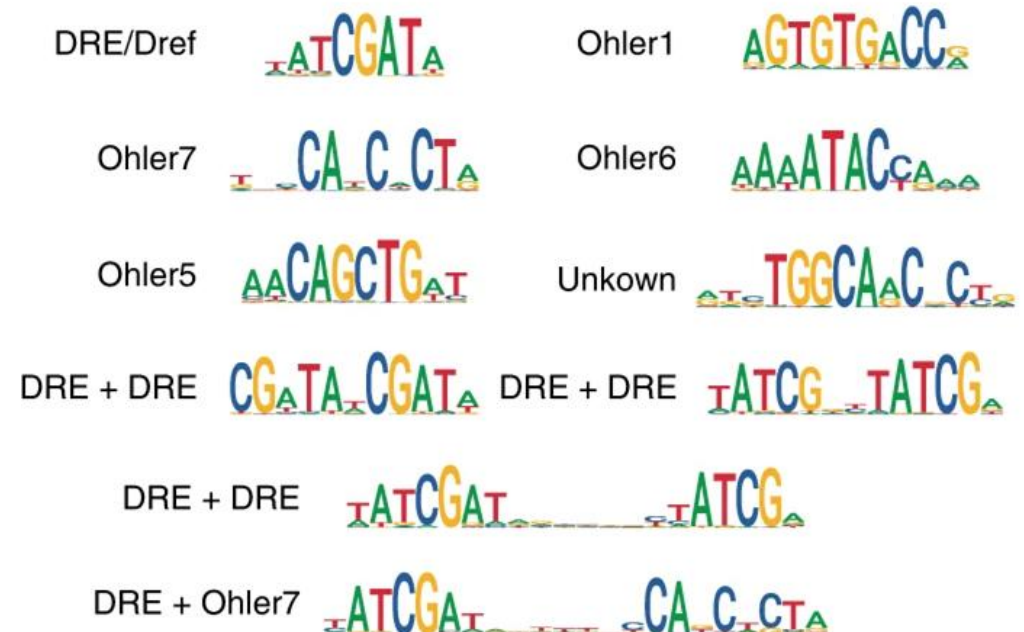
<https://xkcd.com/925/>

Example: DeepSTARR

Dev TF motifs (DeepSTARR)

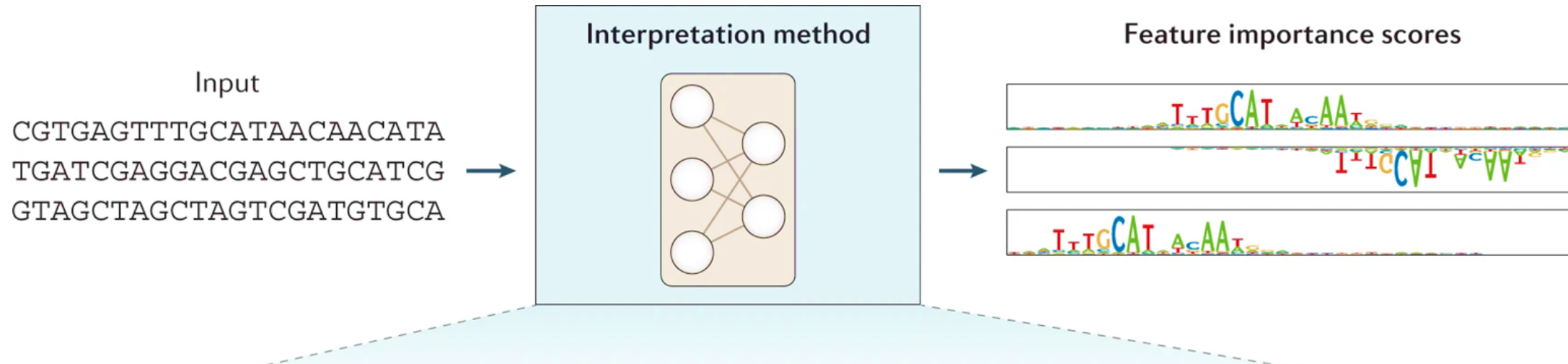


Hk TF motifs (DeepSTARR)



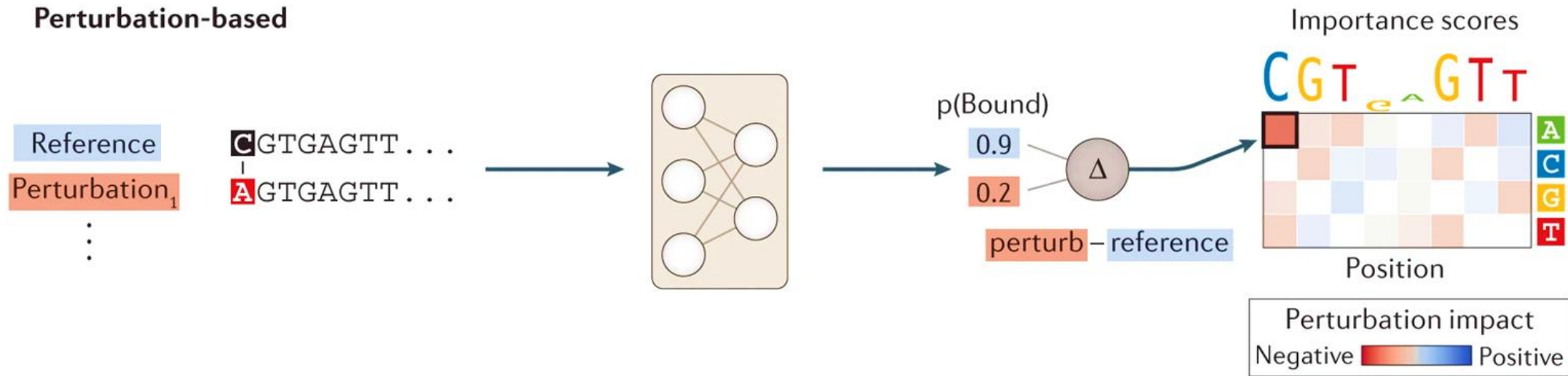
de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

Model Interpretation - the idea



Model Interpretation - ISM

Perturbation-based



Model Interpretation - ISM

Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks

David R. Kelley¹, Jasper Snoek² and John L. Rinn¹

[+](#) Author Affiliations

Corresponding author: dkelley@fas.harvard.edu



Hands-on: Interpreting models with in-silico mutagenesis



https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

Go to: 05_sequence_models/02_ism_and_tfmodisco.ipynb

Klick: Open in Colab

Acknowledgements

- Pyaree M. Dash (BIH – Berlin, Germany)
- Thorben Maaß (UKSH – Lübeck, Germany)

CONTACT

Arjun Devadas arjun.devadas@uni-luebeck.de
Max Schubach max.schubach@bih-charite.de



kircherlab.github.io



UNIVERSITÄT ZU LÜBECK



UNIVERSITÄTSKLINIKUM
Schleswig-Holstein

BIH Berlin Institute
of Health
@Charité

The background of the image is a dense field of coffee beans, likely Arabica, which are dark brown and have a glossy, slightly wrinkled texture. The beans are scattered across the entire frame, creating a rich, textured backdrop. The lighting is soft and even, highlighting the natural curves and creases of the beans.

COFFEE

BREAK

